

Supplementary Information

Landscape-scale bacterial biogeography across the Rub' al-Khali

S1 Analysis provenance

The grouped ecology, rainfall, spatial, XRF, network and encoded-function modules reported in the manuscript write machine-readable TSV and JSON results under `analysis/v3` and `analysis/xrf_audit`. Input manifests, parameter files and checksums are present only for the modules that emit them. Seeds were 20260723 for the core paired, XRF-rescue, spatial-turnover, phylogenetic and functional-redundancy analyses; 20260725 for compartment composition, depth/extraction and XRF CLR; and 20260728 for geographic prediction, spatial-resolution and distance-decay analyses. The post-hoc rainfall pulse analysis used seed 20260804 and 19,999 conditional lag rotations for each model. Its search covered 60 candidate peak times and three diversity endpoints. Sampling and peak-selection intervals used 9,999 whole-site bootstrap samples with seed 20260805. The cross-desert context analysis used seed 20260804, 9,999 depth-zone label permutations and 100 rarefactions per Atacama pit profile. The descriptive taxon-context module used seed 20260903 and 20 subsamples of 8,000 reads per profile for the genus-set overlaps. The biology-context and trait-gene modules used seed 20260904, 9,999 permutations or sign flips and 2,000 site bootstraps. The site-level climate analysis declares separate Benjamini–Hochberg families for nine diversity and 600 genus tests. Its diversity intervals used 9,999 paired-site bootstrap samples with seed 20260805. The PICRUST2 ecology analysis used seed 20260804, 9,999 whole-site permutations and 2,000 site bootstraps; its three primary pathway-profile contrasts, 600 paired pathway tests and 200 geographic pathway tests form separate declared families. The spatial and distance-decay modules use delete-one-site jackknife intervals; every pair involving the omitted site is removed in pairwise analyses. The archived-soil pH module uses dataset version EQ-PH-SHARED-v1.0.0, 999 permutations with seed 20260802, an exact input manifest, recorded software versions and output checksums under `analysis/v3/ph_shared_v1/`. The primary paired and evenness tables use the same 1,000,000-resample bootstrap implementation and seed 20260723; their estimates, intervals and unadjusted p values for the identical Shannon estimand agree. Their q values differ because the primary Shannon family contains three all-campaign plus 14 campaign-specific contrasts, whereas the evenness module corrects three all-campaign contrasts. Printed Shannon intervals come from `analysis/v3/results/paired_compartment_effects.tsv`; the evenness rerun must reproduce them. Multiplicity correction was applied within each declared family; the diversity, composition, rainfall and XRF analyses were corrected separately. The network parameter file records its base and phase seeds. PMA mean-difference intervals used 99,999 paired-aliquot bootstrap samples and encoded-function agreement used 9,999 whole-site bootstrap samples, both with seed 20260805. A typed summary of 44 headline estimates, methods and source results is archived under `analysis/v3/headline_uncertainty/`. Table S1 distinguishes the observational units and cohorts used by each analysis.

S2 Data layers and denominator reconciliation

Table S1: Current analysis denominators. Each denominator corresponds to one data product.

Data product	<i>n</i>	Analytical role
Observed ecology site labels	64	Numeric site labels represented in the alpha table
Primary repeated-site cohort	60 sites; 1,227 profiles	Sites 1 – 60, aggregated to 633 site-campaign-position observations
QC-passed alpha-diversity profiles	1,237	Biological profiles with at least 1,000 reads; 1,227 profiles belong to the 60-site repeated-inference cohort
Quality-controlled profiles per expedition	T1 325; T2 24; T3 478; T4 177; T5 233	The 1,237 QC-passed profiles by expedition (surface, shallow-subsurface, root-adjacent: T1 113, 108, 104; T2 6, 7, 11; T3 168, 148, 162; T4 60, 59, 58; T5 30, 60, 143), as shown in main-text Figure 1b
Canonical all-trip feature table	1,271	Biological and control profiles; source for the 351,472-ASV feature universe
Complete field-XRF sessions	71 of 106 log entries. Trip 5 in-situ handheld measurements retained at 58 sites from a log spanning 59 sites; the log links TestID to site only	
Laboratory-XRF analytical records	725	547 records from Trips 1 – 4 plus 178 Trip 5 records; all included in the canonical elemental-axis analysis
Laboratory-XRF/alpha joins	622	Site-campaign-compartment observations on the 60 core sites
Laboratory-XRF/composition joins	621	Bray-Curtis community analysis after minimum-count filtering
Archived-soil pH/ecology joins	562 alpha; 558 composition	Quality-controlled, incomplete shared archived-sample pH cohort (EQ-PH-SHARED-v1.0.0)
Matched network cohort	170 per compartment	Same campaign-site groups in surface, shallow-subsurface, and root-adjacent soil; 60 sites
Resolution-matched encoded-function null	150	27 surface, 44 shallow subsurface and 79 root-adjacent profiles; 990 annotated genomes, 2,000 retained KOs and 28.5 % mean annotated-catalog fraction
PICRUSt2-shotgun overlap PICRUSt2 ecology cohort	125 1,227 profiles; 633 grouped profiles	Whole-KO-profile comparison Sixty core sites, 462 predicted MetaCyc pathways and 172 complete three-position site-campaign blocks
Retained CBB-marker genome records	132	RuBisCO plus phosphoribulokinase annotations; the complete upstream screened-genome manifest was not preserved, so no denominator or fraction is reported

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Table S1 continued

Data product	<i>n</i>	Analytical role
Rainfall pulse cohort	617 site-campaign- position groups	Technical means for 196 surface, 203 shallow-subsurface, and 218 root-adjacent groups across 60 sites; the paired bulk-soil sensitivity used 179 pairs; 20 distinct NASA POWER daily precipitation series
PMA pairs	9	Trip 5 treated-versus-untreated aliquots

The PMA ASV count table, 150 CoverM profiles, and eggNOG-derived annotations used by the downstream analyses are enumerated with paths and checksums in the analysis manifests under **analysis/v3/**. Reproducing the upstream PMA and shotgun processing additionally requires the corresponding raw-read accessions, dereplicated MAG FASTAs and processing metadata.

The final canonical **feature-table-trips1-5.tsv** comprised 1,271 profiles and 351,472 ASVs. The ecological dataset excluded 24 extraction-blank profiles and ten biological profiles with fewer than 1,000 reads, leaving 1,237 quality-controlled biological profiles. Of these, 1,227 profiles from sites 1 – 60 formed the repeated-inference cohort. Ten profiles from four Trip 1-only locations were retained for description and excluded from repeated-site inference. The primary ecological analysis applied no rare-feature or contaminant filter. Hurlbert expected richness was standardized at 25,000 reads. Sixty of the 1,237 profiles, including 55 of the 1,227 core profiles, fell below that depth, leaving 1,177 full-cohort and 1,172 core expected-richness values. Shannon diversity remained available at full depth for every profile.

S2.1 Wet-laboratory and amplicon-processing record

Each field replicate was placed in a sterile tube by scooping directly with the tube or with a sterile plastic spoon. Samples were held at -20°C in an Engel MR040F portable freezer during each expedition and transferred to -80°C in the laboratory. DNA extraction used 0.9–1.2 g of soil, the QIAGEN PowerLyzer PowerSoil kit (catalogue 12855–50) or DNeasy PowerSoil Pro kit (catalogue 47014), PowerLyzer 24 homogenization, manual or QIACube Connect processing, and a $50\mu\text{L}$ elution.

The Bakt_341F and Bakt_785R primers carried Illumina overhang adapters. Primary amplification used the QIAGEN Multiplex PCR Kit (catalogue 206143): 95°C for 15 min; 35 cycles of 94°C for 30 s, 57°C for 90 s and 72°C for 90 s; and a final 72°C extension for 10 min. Amplicons were checked on a 2% agarose gel, indexed and cleaned following the Illumina 16S library protocol, quantified with the Qubit high-sensitivity double-stranded DNA assay, sized on an Agilent TapeStation 4200, pooled by concentration and size, and sequenced as paired 2×250 -base reads on a NovaSeq 6000.

The canonical feature table was generated with version v2.14.0 of the nf-core Ampliseq workflow. It used DADA2 and SILVA 138.2. The exact command, resource configuration and portable samplesheet are in **data/processed/amplicon/**. The retained laboratory record omits the PCR template input mass, exact index-kit and adapter part numbers, cleanup and indexing cycle counts, NovaSeq reagent kit, and container digest; these items remain unknown.

S2.2 Assay-aware control analysis

The 24 control profiles in the canonical table all come from the Trip 5 sequencing run: EB1–EB18 are extraction blanks and Negative1, Negative2, and Negative4–Negative7 are no-template PCR

controls (water in place of template). For Trips 4 and 5 one blank was processed per extraction day, and extraction days were trip-specific. For Trips 1–3 the laboratory processed one extraction blank per extraction kit (one for PowerLyzer PowerSoil and one for DNeasy PowerSoil Pro); extractions were repeated, Trips 1 and 2 were extracted on the same days, and nothing was sequenced on an extraction day, so no day-linked blank exists for these trips. Of the two kit blanks only the PowerSoil Pro blank was sequenced (**Extraction-Ctrl-Pro-Trip1**, library M-25-0929); no PowerSoil blank appears on any sequencing sample sheet. Among the sequenced Trips 1–3 profiles the recorded kit was PowerSoil for 77, 10 and 359 profiles and PowerSoil Pro for 217, 11 and 87 profiles (Trips 1, 2 and 3; 46 profiles have no recorded kit). Extraction blanks are therefore linked to extraction days or kits. The six Trip 4 extraction blanks (SEB, EB1–EB5) were sequenced on a separate run, outside the canonical table, and were denoised from the raw reads with the same nf-core/ampliseq settings for the Trip 4 screen below. The extraction-day workbook maps EB1–EB17 to dates and 220 Trip 5 biological profiles, of which 217 occur in the canonical table. One EB can therefore be linked to profiles from more than one trip. EB18 and the six **Negative**-labelled profiles lack extraction-day mappings and were retained for characterization only.

The primary screen pooled EB1–EB17 because there was only one blank per extraction batch. It used an uncorrected one-sided Fisher $p < 0.1$ and required presence in at least two blanks. Positive controls were excluded from training. The screen identified 351 of 351,472 ASVs; at the same threshold, relaxing the minimum to one blank called 12,051 ASVs. Those ASVs were removed only from the 217 biological profiles linked to the mapped Trip 5 batches; the other 1,020 ecological profiles were unchanged, and the unfiltered canonical table was retained. Removal accounted for 2.1866 % of reads across the mapped profiles, with median 0.4027 % and maximum 56.5951 % per profile. No profile fell below 25,000 reads. Shannon diversity before and after filtering had Spearman $\rho = 0.99496$ (median change -0.00340 ; maximum absolute change 0.67805), and raw richness had $\rho = 0.99995$ (median change -11 ASVs; maximum absolute change 81). All 25 tracked headline metrics retained their scientific verdicts. Control DNA concentrations were unavailable, so the screen used prevalence only.

A second screen used the six Trip 4 extraction blanks against the 95 Trip 4 profiles linked through the corrected extraction-batch workbook (28 of the 60 Trip 4 sites have no sequenced blank). With the same thresholds it identified 7 candidate ASVs (four unassigned at genus level, *Nesterenkonia*, *Brevibacterium* and *Alcaligenes*); 31 of the 95 profiles lost at least one read, with median removal 0.00 % (interquartile range 0.00–0.01 %), maximum 14.31 % and 0.14 % pooled, and no profile fell below 25,000 reads. Rerunning the 25 tracked headline metrics with both screens applied together (312 filtered profiles) retained every verdict; the largest shifts were the paired Shannon q (0.0256 to 0.0312) and the paired distance-decay omnibus p (0.0053 to 0.0075).

The positive-control design was trip-specific. Trips 1 and 2 used purified-DNA ZymoBIOMICS HMW DNA Standard D6322, which evaluates PCR, library and sequencing performance and enters after extraction and lysis. Trip 3 used whole-cell ZymoBIOMICS Microbial Community Standard D6300 and therefore included extraction. Trip 4 had no positive control. Trip 5 also used D6300, but its positive and negative control pair was sequenced by shotgun metagenomics rather than 16S. Four positive-labelled profiles in the pre-decontamination 16S table, two from Trip 2 and two from Trip 3, were used only for evaluation. At genus level, the two Trip 2 profiles recovered seven and zero of seven expected bacterial genera, and the two Trip 3 profiles recovered eight of eight. Manufacturer genomic-DNA fractions are composition specifications. Under the trip-design product hypotheses, the fractions of all reads assigned to expected bacterial genera were 31.05, 0, 100 and 99.99 % for **e0872_TMC1**, **e0874_Positive**, **e8661_FPosCtrl2** and **e8665_FPosCtrl3**, respectively. Only the last two met the declared descriptive mock-dominated threshold of at least 95 %. These library-to-product assignments remain provisional. Exact mock-associated ASVs were

also traced into negative controls, neighbouring indices and biological profiles; natural occurrence, cross-contamination, index hopping and a mislabelled source library remain open explanations for the overlap. Trip 1 has no positive control with a confirmed identity, and its numbered control libraries were excluded.

The audit follows the principle that all available processing controls should be sequenced and reported, with positive controls reserved for evaluating expected recovery [7]. Three paired 16S libraries with explicit PCR or NTC labels (**e0555_PCR_Ctrl_Trip1**, a nuclease-free water control of Trip 1; **e0875_NTC_2**, the Trip 2 no-template control; and **e8667_PCRCtrl**, the single Trip 3 PCR blank, index pair 728, sequenced on the Trip 3 run with all Trip 3 libraries) and one Trip 4 workbook record bear PCR-blank roles in the control registry. The Trip 3 run (NovaSeq 6000 A01018 run 321, flowcell HVJY5DRX3) also carried three UV-treated sterile-soil negatives (**e8672_FNegCtrl1**, **e8676_FNegCtrl2** and **e8649_NegCtrl3**, the third member of the set, which kept its raw label) and the positive standards. The reaction or plate identifiers of the PCR controls and the biological libraries amplified with them were not recorded, so PCR blanks remain unlinked to extraction batches.

The one sequenced Trips 1–3 extraction blank, **e0929_Extraction_Ctrl_Pro_Trip1** (the PowerSoil Pro kit blank), was excluded: re-denoised from the raw reads, it held 408,918 reads in 1,165 ASVs with 84 % of reads in ASVs shared with biological profiles, a soil profile. It lies at Bray–Curtis 0.39 from a single Trip 1 profile, within the range of same-site replicate pairs, but at 0.93–0.98 from the four libraries prepared alongside it, which points to an isolated transfer into the blank. It was excluded from screening, so Trips 1–3 have no usable sequenced extraction blank.

Because Trips 1–3 have no usable blank, we tested directly whether co-processed profiles share DNA. Within each library series we ranked profiles by library number, called pairs prepared together when their ranks differed by at most two, and compared the Bray–Curtis dissimilarity of these pairs with that of pairs from different sites at the same geographic distance decile and compartment match; the difference was tested by permuting library order 999 times. Co-prepared pairs were as similar as matched pairs in Trip 1 (318 pairs among 325 profiles; difference -0.003 , one-sided $p = 0.15$) or Trip 3 (716 pairs among 449 profiles; $+0.007$, $p = 1.0$), and profiles with low DNA yield, which cross-contamination would affect most, were as close to their neighbours as high-yield profiles (Spearman correlation between log yield and neighbour excess similarity 0.11 and 0.08; $p = 0.10$ and 0.12). Trip 4 showed no excess either ($+0.016$, $p = 1.0$), whereas Trip 5, which had per-day blanks and was screened, showed a small excess (-0.022 , $p = 0.001$), consistent with the batch signal that the Trip 5 screen removed. The transfer into the Trips 1–3 blank therefore left no detectable trace among the biological profiles.

The material limits are that Trips 1–3 have no usable extraction blank; that the Trip 4 blanks of three extraction days were not sequenced, affecting all or part of 28 sites, and that the Trip 4 PCR blank and no-template control were not sequenced; that Trip 5 has no positive control in the 16S assay; and that no sterile-bag inventory was recovered.

S2.3 Two XRF measurement workflows

Table S2 separates the coverage, material and analytical use of the field and laboratory workflows.

Table S2: Field and laboratory XRF are distinct measurement workflows.

Workflow	Coverage	Material	Analytical use
Field XRF	Trip 5 only; 106 log entries across 59 sites, of which 71 complete sessions at 58 sites were retained	In situ Vanta handheld sessions; TestID–site provenance only	Field-scale context and descriptive Trip 5 site-level diagnostic
Laboratory XRF	All five trips; 725 records; 622 alpha and 621 composition joins	Archived collected samples measured in the laboratory	Within-campaign elemental PCA and adjusted ecology models; Trip 5 settings encoded, Trips 1 – 4 method/mode and combined-source units/LOD semantics absent

The source-specific analytical tables reproduce 547 Trips 1 – 4 records and 178 Trip 5 records. The canonical analysis includes all 725, keeps field XRF separate, excludes Trip-1-only sites 61 – 64 from repeated-campaign inference and reduces between-campaign location differences by standardizing elements within campaign. Method, mode, material and diameter are encoded for Trip 5 laboratory records but absent from the consolidated Trips 1 – 4 source; concentration units, limits of detection and non-detect semantics are not machine-readable across the combined source. The Trip 5 field–laboratory comparison is a site-level diagnostic because the field log contains no physical-aliquot identifier. Trips 1 – 4 used the maximum positive analyte repeat and Trip 5 retained the last repeat. A primary-status sensitivity for the 11 affected Trip 5 analytes gave Spearman $\rho \geq 0.994$, so this bookkeeping difference left the analysis ordering intact.

Instrument zeros were treated as non-detects. Eleven elements detected in at least 75 % of laboratory specimens (Al, Ca, Cl, Fe, K, Mg, Na, P, S, Si and Ti) were transformed as $\log(1 + x)$, standardized within campaign and submitted to PCA. PC1 explained 39.2 % of variance; its largest positive loadings were Ca (0.413), Mg (0.406), Na (0.370), S (0.354) and Cl (0.338), with Si negative (–0.273). PC1 correlated $r = 0.825$ with an internal evaporite-element reference and $r = 0.976$ with the globally standardized sensitivity axis.

The composition of the axis supports the descriptive term *evaporite-associated elemental axis*. In the site-fixed alpha model ($n = 622$), the Shannon coefficient was 0.00027 (95 % CI [–0.040, 0.041], $p = 0.990$), so no diversity association was detected. In the primary Bray–Curtis model ($n = 621$), 38 PCoA axes retained 80.0 % of variation and the elemental term had partial $R^2 = 0.00345$ ($p = 0.056$, 999 within-site permutations); retaining 95.0 % of variation gave a similar $R^2 = 0.00330$ but $p = 0.044$. The estimated effect size was therefore stable across that threshold while the permutation p value moved from 0.010 to 0.044.

Because amplicon composition is compositional, a prespecified Aitchison sensitivity reused the same 621 joins and the unchanged axis. Genera present in at least 20 % of groups were ranked by mean relative abundance, and the nine combinations of 80, 200 or all 383 eligible genera with a 0.5 or 1.0 pseudocount or multiplicative zero replacement at 0.65 all supported the axis after campaign, compartment and site adjustment (partial $R^2 = 0.00311$ –0.00543, $p = 0.002$ –0.015; primary $R^2 = 0.00358$, $p = 0.010$). The primary configuration used the top 200 genera and a 0.5 pseudocount. The warranted conclusion is a small conditional composition association, identified within sites. A separate diagnostic that omitted site indicators, because they force site-mean residuals to zero, found that site-mean CLR residuals remained spatially autocorrelated after campaign, compartment

and elemental-axis adjustment (multivariate Moran’s $I = 0.304$, $p = 0.001$), so broad geographic structure remains after elemental-axis adjustment.

XRF measures total elemental composition. Because the analysed cohort has no independent electrical-conductivity measurement, we call the gradient an “evaporite-associated elemental axis”.

S3 Archived-soil pH sensitivity

The analysis used archived-soil pH dataset version EQ-PH-SHARED-v1.0.0. We measured pH in 767 specimens and retained 712 measurements with an unambiguous measurement date, a value in the physical range, recorded calibration, an electrode slope of $95 - -102\%$, and a pH 10 buffer read-back of $9.9 - -10.1$. 36 measurements had ambiguous dates and 19 failed session quality-control criteria. Analyses used admitted measurements only. Coverage was incomplete and non-random, especially in campaigns 1 and 3 (Table S3).

Table S3: Coverage and quality control of the archived-soil pH measurements.

Campaign	Target specimens	Measured	Passed QC	Not measured	Excluded
1	330	157	157	173	0
2	27	23	23	4	0
3	450	192	156	258	36
4	177	175	156	2	19
5	234	220	220	14	0

The versioned SIO graph models each retained measurement as an existing specimen, a PATO acidity quality (PATO:0001842), a value typed as both an SIO measurement value (SIO:000070) and pH (SIO:001089), and an SIO measuring process (SIO:001054). Values use the pH unit (UO:0000196); process, value, quality, protocol, device and session links follow the same directions as the XRF and DNA measurement patterns. Stable IRIs depend on specimen, date and protocol. The shared version contains 712 process–quality–value sets and 29 reconstructed session groupings based on trip, date, slope and pH-10 read-back. The source workbook does not contain a session identifier. A deposited shape map enumerates every validated process, value, quality, protocol, device and session; a deposited negative fixture without the pH unit is rejected. Graph-equivalent Turtle and RDF/XML serializations are imported by the canonical knowledge graph.

The ecology join contained 702 specimens, 562 alpha-diversity groups and 558 composition groups across all 60 core sites. Table S4 reports the model results. Availability is non-random, especially within campaigns 1 and 3, so inference is bounded to this subset.

Table S4: Archived-soil pH sensitivity results for EQ-PH-SHARED-v1.0.0.

Endpoint	n	Effect size	p	Boundary
Site-fixed Shannon	562	$\beta = 0.185$	0.483	No detected diversity association
Primary top-200 CLR composition	558	partial $R^2 = 0.00274$	0.126	No detected direct association
Bray–Curtis composition, 80 % axes	558	partial $R^2 = 0.00261$	0.275	No detected direct association
Transect, same cohort before pH	60 sites	partial $R^2 = 0.358$	≤ 0.001	57 residual df; same-cohort reference
Transect, same cohort after pH	60 sites	partial $R^2 = 0.196$	≤ 0.001	57 residual df; attenuated
pH trend along transect	60 sites	partial $R^2 = 0.846$	≤ 0.001	57 residual df; archived subset

The pH maximum was the Rhizosphere group from campaign 4, site 60, based on 1 specimen. Excluding that group changed the pH transect partial R^2 from 0.846 to 0.826 and the pH-adjusted composition statistic from 0.196 to 0.187. Excluding all of site 60 gave corresponding values of 0.833 and 0.188; all refit permutation tests remained at $p \leq 0.001$. The maximum therefore influences the descriptive pH gradient while the paper’s conclusion holds without it.

Across 60 sites with all 3 positions, the pH-position omnibus gave $\chi^2 = 19.757$ ($p = 5.13 \times 10^{-5}$). The Wilcoxon contrasts used 60/60, 60/60 and 59/60 nonzero/site differences for shallow-subsurface–surface, root-adjacent–surface and root-adjacent–shallow-subsurface, respectively. The differences are small; the ± 0.1 pH-10 read-back quality-control band is a session quality-control criterion.

S4 Evidence status

Table S5 links each central result to its supported interpretation.

Table S5: Claim ledger.

Topic	Evidence	Supported interpretation
Soil-position composition	170 complete campaign–site blocks; omnibus pseudo- $F = 17.28$, $p = 0.001$; pairwise-complete cohorts of 183, 182 and 191 blocks gave three CLR displacements with $p = q = 0.001$; surface-to-root-adjacent dispersion increased, but the final shallow-to-root step had $q = 0.061$	Centroid location and spread differ among operational positions

Topic	Evidence	Supported interpretation
Root proximity	Campaign-averaged, site-paired Shannon: root-adjacent-surface mean -0.130 [95 % CI -0.321 , 0.079], $q = 0.0256$; root-adjacent-shallow mean -0.206 [-0.398 , -0.001], $q = 0.0256$, with the bootstrap upper endpoint adjacent to zero; post-hoc normalized-evenness contrasts -0.0315 and -0.0440 , expected-richness contrasts $q \geq 0.0675$	Lower root-adjacent Shannon accompanies lower normalized evenness
Geography	Centred-CLR joint linear-plus-quadratic transect model partial $R^2 = 0.4007$, $p = 0.001$; campaign-plus-block holdout $R^2 = -0.0017$; collection-order $ \rho = 0.994$ – 1.000 ; residual Moran's I depends on k	Repeatable transect structure within the surveyed campaigns; the shared pH cohort below accounts for part of the same-cohort statistic
Archived-soil pH	702 exact specimen joins; 558 composition groups; direct primary CLR partial $R^2 = 0.00274$, $p = 0.126$; same-cohort transect partial R^2 changed from 0.358 to 0.196 after pH adjustment	pH is a measured correlate of part of the spatial pattern
Assembly	Top-800-ASV laboratory-XRF elemental-axis niche diagnostic: overall Mantel $r = -0.0837$ and closest-30 % pairwise $r = 0.0262$	Short-distance phylogenetic signal was absent, so β NTI/RC-Bray process categories are withheld
Elemental axis	Within-campaign PC1 explains 39.2 %; adjusted Shannon $p = 0.990$; primary CLR composition partial $R^2 = 0.00358$, $p = 0.010$; all nine CLR configurations span $0.31 - -0.54$ %	Descriptive evaporite-associated elemental gradient; a small conditional composition association, with broad geographic structure remaining after adjustment
Long-term climate	Forty-nine months at each of 60 sites; all nine temperature/rain/humidity associations with Shannon, expected richness and evenness pass global correction; 111 – -112 of 200 genus tests per climate summary pass a 600-test correction	Climate gradients coincide with bacterial diversity and abundance gradients and covary closely with the transect
Campaign/depth	Campaign-by-position Wald $p = 0.00831$ unadjusted and $p = 0.17476$ after log-sequencing-depth adjustment; kit recorded for 858/1,237 profiles; QIACube occurs only in campaigns 4 and 5, whereas campaigns 1 – 3 contain multiple recorded kits	Campaign-by-position interaction is depth-sensitive; laboratory batch effects remain in the extraction fits

Topic	Evidence	Supported interpretation
Rain response	Rise-and-decay model across 617 site-campaign-position groups, 60 candidate peaks and 3 endpoints; 19,999 conditional lag rotations; expected richness peaked at 2 days with NASA POWER ($p = 0.0560$) and at 1 day with Open-Meteo ($p = 0.0432$) after directional family-wise correction	Evidence for a short richness association during the first several days
PMA	9 Trip 5 pairs; Hurlbert expected richness lower after treatment, no Shannon difference detected	Direct paired endpoints for nine Trip 5 pairs
Contamination	Assay-aware screen: 17 extraction-day blanks with batch links to Trip 5 profiles; 351/351,472 candidate ASVs; 2.19% of reads removed across 217 mapped profiles; four 16S positive profiles evaluated; 25 headline metrics rerun	All headline verdicts stable under the filter, which is trained on the Trip 4 and 5 profiles with linked blanks
Conditional networks	170 matched observations per compartment; 80 genera; CLR Graphical Lasso; 200 site-cluster bootstraps; null/raw edge ratio 0.05374	Stable null-calibrated conditional associations occur within each compartment; density ordering among compartments varies across bootstraps and configurations
Encoded-functional turnover	150 analysis-ready metagenome profiles; 990 annotated genomes; 2,000 KOs; observed 0.13386 versus taxonomic 0.82272 and null median 0.08733, lower-tail $p = 1.0$, upper-tail $p = 0.001$	Lower raw encoded-functional than taxonomic turnover, but higher than the resolution-matched genome-label null
PICRUSt2	1,227 core profiles; pathway-composition transect $R^2 = 0.234$, $p = 0.0001$; 172 complete position blocks; 125 shotgun matches with whole-KO median $\rho = 0.74$; RuBisCO $r = 0.16$, $p = 0.067$	Broad predicted potential follows geography and soil position and agrees with shotgun at whole-profile scale
Carbon markers	132 genome records with RuBisCO+PRK annotations; complete upstream screened-genome denominator unavailable	Candidate Calvin-cycle genomic potential in 132 genome records
Trace-gas markers	Product/KO annotation screen without catalytic, phylogenetic, synteny or direct-read confirmation	Candidate marker annotations awaiting catalytic, phylogenetic and synteny confirmation

S5 Compartment and spatial evidence

S5.1 Paired compartment composition

The primary composition analysis aggregated genus counts within campaign-site-position and retained 170 complete three-position blocks across 60 sites for the omnibus test. Genera present in at least 20% of groups were ranked by mean relative abundance; the primary configuration used the leading 200 genera, a 0.5 pseudocount and centred-log-ratio (CLR) coordinates. Permuting

position labels within blocks gave an omnibus pseudo- $F = 17.2797$ with $p = 0.001$.

Pairwise location tests used their respective pairwise-complete cohorts and sign-flipped the mean within-block CLR difference vector across sites. Standardized displacements were 0.3312 for shallow-subsurface minus surface ($n = 183$ blocks), 0.4328 for root-adjacent minus surface ($n = 182$) and 0.4780 for root-adjacent minus shallow-subsurface ($n = 191$); each had permutation $p = 0.001$ and Benjamini–Hochberg $q = 0.001$. The directions and location results persisted with 80, 200 or all 384 eligible genera, three zero treatments and each leave-one-campaign refit. Because the test targets the paired mean difference vector, it is exact under the paired-symmetry null even when within-position dispersion differs.

Dispersion increased monotonically from surface to shallow-subsurface to root-adjacent soil. The shallow-subsurface–surface difference was 1.3776 (bootstrap 95 % CI [0.4411, 2.3748], permutation $q = 0.015$); root-adjacent–surface was 2.3866 ([1.5502, 3.2123], $q = 0.003$); and root-adjacent–shallow-subsurface was 1.0090 ([0.0556, 2.0550], $q = 0.061$). The last permutation contrast has $q = 0.061$ with a bootstrap interval excluding zero. The composition result describes the operational collection position.

Testing every one of the 200 genera in the same block-centred site tensor (three paired contrasts, 600 sign-flip tests, one Benjamini–Hochberg family) supported 295 differences: 72 in the shallow-subsurface–surface contrast (33 higher below the surface, 39 higher at the surface), 109 in the root-adjacent–surface contrast (54 and 55) and 114 in the root-adjacent–shallow-subsurface contrast (53 and 61). The per-genus means reproduce the displacement loadings of Figure 2d exactly. Table S16 lists the leading genera per contrast and direction.

S5.2 Alpha diversity and post-hoc normalized evenness

For all-campaign paired contrasts, profiles were averaged within site–campaign–position and then across campaigns within site and position. Among 60 sites, root-adjacent minus surface Shannon diversity had mean -0.130 (median -0.250 , bootstrap 95 % CI $[-0.321, 0.079]$, Wilcoxon $q = 0.0256$). Root-adjacent minus shallow-subsurface had mean -0.206 (median -0.430 , 95 % bootstrap CI $[-0.398, -0.001]$, upper endpoint adjacent to zero, $q = 0.0256$). The distributional claim is carried by the paired Wilcoxon result. shallow-subsurface minus surface had mean 0.075 (95 % CI $[-0.089, 0.235]$, $q = 0.189$).

A post-hoc decomposition used $H/\log(E[S_{25k}])$, where $E[S_{25k}]$ is Hurlbert expected richness at 25,000 reads. This is a normalized-evenness sensitivity. Root-adjacent minus surface normalized evenness was -0.03148 (95 % CI $[-0.04477, -0.01863]$, $q = 2.76 \times 10^{-6}$), root-adjacent minus shallow-subsurface was -0.04398 ($[-0.05717, -0.03131]$, $q = 2.13 \times 10^{-7}$), and shallow subsurface minus surface was $+0.01250$ ($[0.00084, 0.02508]$, $q = 0.0121$). Campaign- and log-depth-adjusted GEE contrasts retained these directions: -0.02841 ($p = 1.67 \times 10^{-5}$), -0.04260 ($p = 3.24 \times 10^{-11}$) and $+0.01419$ ($p = 0.0125$), respectively. Expected-richness contrasts gave $q = 0.814$, 0.0878 and 0.0675 for root-adjacent–surface, root-adjacent–shallow and shallow–surface after their three-test correction. Hurlbert richness was unavailable for 55 core profiles below 25,000 reads, and normalized evenness was also undefined for one single-ASV profile; its GEE used 617 of 633 blocks.

S5.3 Transect structure, transport and spatial resolution

The spatial analysis retained 630 site–campaign–position groups from the 60 core sites after a 2,000-read threshold. Genus counts were CLR-transformed, centred within campaign and position, and averaged by site. With the 200 most common eligible genera, the joint linear-plus-quadratic transect trend explained partial $R^2 = 0.4007$ ($p = 0.001$, 999 site-level permutations). Linear

and quadratic models using 80, 200 or all 384 eligible genera each gave $p = 0.001$, as did all five leave-one-campaign-out fits. These are influence analyses: campaign shares of the 630 groups were 25.2 %, 2.5 %, 26.8 %, 27.9 % and 17.5 %, with the smallest campaign covering eight sites at one end of the route.

Elapsed collection time and transect position were nearly perfectly aliased: Spearman $\rho = 1.000$, 1.000, 0.9938, 1.000 and 1.000 for campaigns 1 – 5. The primary transport arm selected taxa within each training fold. It held out campaigns 1, 3 and 4, the campaigns with complete 60-site traversals, in turn with a contiguous ten-site block. Equal-weight $R^2 = -0.0017$, pooled $R^2 = -0.0005$, and 10 of 18 folds had positive skill; whole-site relabelling gave $p = 0.406$ and cyclic shift/reflection $p = 0.083$. Cross-campaign, cross-block transport therefore gave no out-of-sample skill. The group-level sensitivity spanning all five campaigns was also negative ($R^2 = -0.0041$). In a sensitivity that held out a site block while retaining all campaigns on both sides, out-of-block $R^2 = 0.2528$, all six blocks had positive skill, and the same nulls gave $p = 0.002$ and 0.025. This arm tests transport to unseen site blocks within seen campaigns.

At ASV resolution, the same model on the same 630 groups and 60 sites gave partial $R^2 = 0.3728$ (95 % delete-one-site jackknife interval 0.3208–0.4249) with the 800 most abundant ASVs, 0.3538 (0.3043–0.4034) with 2,000 ASVs and 0.3455 (0.2962–0.3948) with all 2,642 eligible ASVs, compared with 0.4007 at genus level; all fits gave $p = 0.001$. One of the 630 groups (campaign 3, site 54, root-adjacent) carried fewer than 2,000 reads on prevalence-filtered ASVs while passing the genus-assigned read threshold; it was retained because the cohort is defined once, by the primary quality control. The genus value above is the primary fit. The transect result therefore holds at ASV resolution, with larger ASV residual Moran’s I (0.149–0.162).

The residual diagnostic depends on the neighbour graph. This independent neighbour-count sweep used seed 20260728 and gave $p = 0.002$ at $k = 5$; the primary fixed- k analysis used seed 20260723 and gave $p = 0.001$ for the same I . Moran’s I declined from 0.1101, 0.1015, 0.0646 and 0.0503 at $k = 3, 4, 5$ and 6 to 0.0070 at $k = 8$ and -0.0098 at $k = 10$. Residual autocorrelation was detected at $k = 3$ to 6 (permutation $p \leq 0.002$) and not at $k = 8$ ($p = 0.050$) or $k = 10$ ($p = 0.243$). Residual spatial structure is therefore confined to short neighbourhoods.

S5.4 Paired distance decay and dissimilarity components

Campaign-averaged position matrices over the same 60 sites yield 1,770 geographic pairs. Whole-site permutations were applied simultaneously to all three matrices while geographic distance remained fixed, so the site was the unit of resampling. Aitchison dissimilarity slopes per 100 km were 1.349 for surface, 1.507 for shallow-subsurface and 1.175 for root-adjacent soil (each $p = 0.0001$; 9,999 permutations). The omnibus position contrast gave $p = 0.0051$. With maximum-statistic family-wise control, shallow-subsurface minus root-adjacent was +0.332 per 100 km ($p_{\text{adj}} = 0.0039$), whereas surface minus root-adjacent was +0.174 ($p_{\text{adj}} = 0.173$).

After coverage standardization to 12,865 reads, mean Sørensen dissimilarity was 0.401, 0.406 and 0.395 in surface, shallow-subsurface and root-adjacent soil. Simpson replacement accounted for 0.286, 0.280 and 0.295, or 69 – 75 % of Sørensen dissimilarity, and both replacement and nestedness components increased with distance under the same permutation maps. Their slopes per 100 km were, respectively, 0.0075 versus 0.0113 in surface soil, 0.0097 versus 0.0165 in shallow-subsurface soil, and 0.0117 versus 0.0055 in root-adjacent soil. Replacement therefore dominated mean dissimilarity in all three positions, while the abundance-gradient component carried the larger share of the distance-related increase in both bulk-soil positions. The unstandardized Bray–Curtis abundance-gradient component correlated with the absolute log library-size ratio at $r = 0.76$ –0.83 and becomes identically zero at common depth, so it is a library-size artefact.

S5.5 Conditional association diagnostic

Thresholded Spearman networks contained 90.6 – –99.9 % positive edges, a warning for compositional and indirect correlation artifacts, and were set aside. For the conditional association diagnostic, sequencing replicates were summed within campaign–site–compartment, aggregate read depth was required to be at least 2,000, and the analysis intersected the three compartments to 170 observations each across 60 sites. Campaign contributions were 42, 2, 53, 56 and 17 observations per compartment. The 80 most abundant genera present in at least 20 % of every compartment were CLR-transformed (pseudocount 0.5), centered within campaign and standardized. A common Graphical-Lasso penalty $\alpha = 0.20$ was selected against a permuted-taxon null; its combined null/raw edge ratio was 0.05374. Stable edges required at least 0.80 selection and 0.90 sign consistency in 200 paired site-cluster bootstraps and null selection frequency at most 0.05. Table S6 reports the resulting stable-edge diagnostics.

Table S6: Stable null-calibrated conditional associations by operational soil position. Expected false fractions are calibration diagnostics.

Compartment	Stable edges	Stable density	Positive fraction	Expected false fraction
Surface	180	0.05696	0.6278	0.0096
shallow-subsurface	221	0.06994	0.5747	0.0087
Root-adjacent	237	0.07500	0.5865	0.0104

Stable null-calibrated conditional associations were therefore recoverable in every compartment. The mean expected false fractions among stable edges are calibration diagnostics. Comparative density ordering varied: 95 % site-cluster bootstrap intervals were $[-0.0142, 0.0089]$ for surface minus shallow-subsurface, $[-0.0161, 0.0070]$ for surface minus root-adjacent and $[-0.0155, 0.0104]$ for shallow-subsurface minus root-adjacent. Density directions persisted in only 0.78, 0.67 and 0.67 of nine taxon-count/penalty configurations. No campaign comparison met the predeclared 80-observation rule, so networks were estimated per compartment only. The retained edges are descriptive regularized conditional associations.

S5.6 Taxon composition and transect replacement

Tables S7 and S8 summarize the phylum, class and genus composition of the 1,227 core-site profiles as mean shares of total reads. Genus sums recomputed from the canonical feature table reproduce the cached genus table used by every inferential module (maximum absolute difference 0 reads). Bacilli (21.0 %), Gammaproteobacteria (14.7 %), Actinobacteria (13.3 %) and Alphaproteobacteria (12.2 %) were the leading classes. *Domibacillus* led every compartment and the western and central thirds; *Bacillus* led the eastern third.

Table S9 lists the genera with the strongest monotonic change along the route among the 200 genera of the route model, in the same site-level CLR coordinates as the geographic model. Of the 200, 124 passed Benjamini–Hochberg $q < 0.05$: 77 decreased and 47 increased eastward. By phylum, the decreasing genera were 27 Actinomycetota, 20 Pseudomonadota, 9 Bacteroidota, 4 Acidobacteriota, 3 Chloroflexota, 3 Myxococcota, 3 Planctomycetota, 2 Gemmatimonadota and 6 others; the increasing genera were 17 Bacillota, 16 Pseudomonadota, 6 Actinomycetota, 4 Planctomycetota, 3 Bacteroidota and 1 Deinococcota. Table S10 gives the direction of the environmental gradients along the same route: temperature, humidity, rainfall and pH all increased from west to east. The correlations describe where genera change.

Table S7: Leading phyla across the 1,227 core-site profiles. Values are mean shares of total reads in percent; prevalence is the fraction of profiles with at least one read. Compartment and transect-third columns are means within the stratum (thirds of 20 sites ordered by route position). Names follow SILVA 138.2.

Phylum	ASVs	Mean	Prev.	Surface	Shallow	Root-adj.	West	East
Pseudomonadota	88,894	26.9	100	29.9	23.8	27.0	26.3	28.2
Bacillota	48,693	22.5	100	20.8	19.6	26.2	18.3	31.5
Actinomycetota	56,581	18.7	99	18.0	20.3	18.1	22.6	12.7
Chloroflexota	24,409	8.6	99	8.5	10.7	6.9	10.3	6.0
Bacteroidota	16,657	7.1	99	8.2	5.9	7.3	6.8	6.0
Planctomycetota	20,808	3.2	98	2.3	3.5	3.8	3.1	2.9
Gemmatimonadota	12,107	2.7	98	2.9	3.5	1.8	2.7	2.8
Acidobacteriota	17,405	2.3	95	2.1	3.4	1.7	2.6	1.7
Myxococcota	15,121	1.7	98	1.6	2.0	1.5	1.7	1.4
Deinococcota	1,419	1.1	93	1.9	1.0	0.6	1.4	1.0

Table S8: Leading genera across the 1,227 core-site profiles (mean share of total reads in percent, prevalence in percent of profiles, and means by compartment and transect third).

Genus	Phylum	Mean	Prev.	Surface	Shallow	Root-adj.	West	Central	East
<i>Domibacillus</i>	Bacillota	6.22	97	6.14	4.09	8.00	7.09	6.33	4.78
<i>Massilia</i>	Pseudomonadota	3.55	94	4.85	2.14	3.66	3.69	4.58	1.97
<i>Bacillus</i>	Bacillota	2.97	98	3.82	2.64	2.58	1.56	2.11	6.22
<i>Microvirga</i>	Pseudomonadota	2.41	97	2.54	2.11	2.55	3.32	1.95	1.66
<i>Flavisolibacter</i>	Bacteroidota	2.05	94	2.88	2.00	1.44	2.49	2.27	1.10
<i>Noviherbaspirillum</i>	Pseudomonadota	1.92	94	2.62	1.76	1.50	2.22	2.10	1.23
<i>Streptomyces</i>	Actinomycetota	1.80	91	1.43	1.91	2.01	2.60	1.71	0.73
<i>Herpetosiphon</i>	Chloroflexota	1.52	92	1.68	2.07	0.95	1.45	2.19	0.72
<i>Paenibacillus</i>	Bacillota	1.38	96	1.08	1.29	1.67	1.22	1.34	1.65
<i>Lysinibacillus</i>	Bacillota	1.32	95	0.65	0.70	2.36	1.57	1.32	0.95
<i>Nocardioidea</i>	Actinomycetota	1.23	95	1.19	1.14	1.34	1.45	1.36	0.74
<i>Ammoniphilus</i>	Bacillota	1.09	96	0.89	1.46	0.95	0.53	1.10	1.92
<i>Nibribacter</i>	Bacteroidota	1.09	84	1.26	0.47	1.46	0.83	1.81	0.53
<i>Planococcus</i>	Bacillota	1.03	82	1.32	0.60	1.14	0.74	0.62	1.99
<i>Deinococcus</i>	Deinococcota	1.02	88	1.83	0.94	0.46	1.39	0.84	0.73

Table S9: Genera with the strongest monotonic change along the route among the 200 genera of the route model. ρ is the Spearman correlation of the site-level CLR value (campaign-by-compartment means removed) with route position; q is Benjamini–Hochberg over 200 tests (77 genera decreased and 47 increased eastward at $q < 0.05$). Δ is the east-minus-west difference in mean CLR between the transect thirds with a delete-one-site jackknife 95 % interval; the last two columns give the mean share of total reads in percent in the western and eastern thirds.

Genus	Phylum	ρ	q	Δ CLR [95 % CI]	West	East
<i>Strongest decreases from west to east</i>						
<i>Cellulomonas</i>	Actinomycetota	-0.79	2.3e-12	-2.23 [-2.86, -1.60]	0.50	0.13
<i>Blastocatella</i>	Acidobacteriota	-0.76	3.8e-11	-2.26 [-2.80, -1.71]	0.10	0.01
<i>Marmoricola</i>	Actinomycetota	-0.70	5.9e-09	-1.65 [-2.26, -1.05]	0.20	0.07
<i>Ellin6055</i>	Pseudomonadota	-0.70	7.9e-09	-1.71 [-2.39, -1.03]	1.12	0.26
<i>Roseisolibacter</i>	Gemmatimonadota	-0.70	9.3e-09	-1.89 [-2.49, -1.28]	0.22	0.09
<i>Steroidobacter</i>	Pseudomonadota	-0.69	1.2e-08	-1.61 [-2.10, -1.12]	0.29	0.11
<i>Limnobacter</i>	Pseudomonadota	-0.69	1.5e-08	-2.83 [-3.50, -2.17]	0.22	0.03
<i>Aridibacter</i>	Acidobacteriota	-0.69	1.5e-08	-1.86 [-2.44, -1.29]	0.08	0.02
<i>Sphingomonas</i>	Pseudomonadota	-0.67	5.5e-08	-1.35 [-1.87, -0.84]	0.21	0.14
<i>Paraconexibacter</i>	Actinomycetota	-0.67	5.5e-08	-1.84 [-2.34, -1.34]	0.05	0.03
<i>Strongest increases from west to east</i>						
<i>Halalkalibacter</i>	Bacillota	0.85	8.4e-16	4.91 [3.83, 5.99]	0.03	0.83
<i>Polygonibacillus</i>	Bacillota	0.81	6.9e-13	4.55 [3.59, 5.51]	0.00	0.20
<i>Ammoniphilus</i>	Bacillota	0.80	9.5e-13	1.89 [1.48, 2.31]	0.53	1.92
<i>Bacillus</i>	Bacillota	0.77	2.5e-11	1.62 [1.18, 2.07]	1.56	6.22
<i>Pseudalkalibacillus</i>	Bacillota	0.74	4.1e-10	2.89 [2.28, 3.50]	0.01	0.05
<i>Aquibacillus</i>	Bacillota	0.72	1.9e-09	4.75 [3.45, 6.04]	0.01	0.69
<i>Plastorhodobacter</i>	Pseudomonadota	0.72	3.0e-09	3.60 [2.74, 4.46]	0.02	0.10
<i>Rhodocista</i>	Pseudomonadota	0.71	5.9e-09	3.01 [2.12, 3.90]	0.03	0.20
<i>Escherichia-Shigella</i>	Pseudomonadota	0.70	7.9e-09	3.27 [2.36, 4.18]	0.01	0.84
<i>Sediminibacillus</i>	Bacillota	0.70	8.7e-09	4.59 [3.46, 5.71]	0.00	0.52

Table S10: Direction of the environmental gradients along the route: Spearman correlation of site means with route position and the means of the western and eastern transect thirds.

Variable	Sites	ρ	West third	East third
49-month mean air temperature	60	0.98	27.89	30.07
49-month mean monthly rainfall	60	0.55	2.16	3.67
49-month mean relative humidity	60	0.92	24.72	31.82
archived-soil pH (admitted measurements, site mean)	60	0.82	7.85	8.23

Table S11: Genus-set overlap after subsampling every profile 20 times to 8,000 reads. A genus counts as detected in a set when it occurs in at least 10 % of subsampled profiles. “Top 50 A in B” is the number of the 50 most abundant genera of set A that are detected in set B. The Atacama pit profiles (PRJEB39249) use V4 primers and intracellular DNA; the comparison is presence-based and does not merge feature tables.

Set A			Set B			Profiles A	Profiles B	Genera A	Genera B	Shared	Jaccard	Top 50 A in B
Empty third	Quarter	west	Empty Quarter	east third		469	309	368	317	267	0.639	
Empty third	Quarter	west	Empty Quarter	central third		469	426	368	364	331	0.825	
Empty third	Quarter	central	Empty Quarter	east third		426	309	364	317	275	0.677	
Empty profiles	Quarter	all core	Atacama pit	all depths (PRJEB39249)		1204	58	364	76	43	0.108	
Empty	Quarter	surface	Atacama pit	2.5-10 cm (PRJEB39249)		368	4	336	61	33	0.091	

Table S12: Predicted pathway classes. Share is the class total of mean relative pathway abundance across the 1,227 core-site profiles (keyword grouping of MetaCyc descriptions; rules archived in `analysis/v3/taxon_context/run_manifest.json`). The supported counts summarize the route-correlation family (200 tests) and the compartment-contrast family (600 tests) of Section S8.1 by class, with the number of positive (eastward increase or first-named compartment higher) and negative estimates.

Class	Pathways	Share (%)	Route (+/-)	Compartment (+/-)
Biosynthesis	231	66.8	63 (54/9)	208 (86/122)
Energy and central metabolism	55	17.2	13 (6/7)	28 (14/14)
Degradation or utilization	135	10.7	10 (1/9)	21 (12/9)
Other	41	5.3	6 (5/1)	13 (5/8)

Table S13: The ten most abundant predicted MetaCyc pathways and the carbon-fixation pathways present in the PICRUST2 pathway set (mean share of predicted pathway abundance in percent; rank among 462 pathways).

Rank	Pathway	Description	Share (%)
1	PWY-3781	aerobic respiration I (cytochrome c)	1.70
2	PWY-7111	pyruvate fermentation to isobutanol (engineered)	1.03
3	PWY-5101	L-isoleucine biosynthesis II	0.93
4	ILEUSYN-PWY	L-isoleucine biosynthesis I (from threonine)	0.91
5	VALSYN-PWY	L-valine biosynthesis	0.91
6	BRANCHED-CHAIN-AA-SYN-PWY	superpathway of branched amino acid biosynthesis	0.80
7	PWY-5973	cis-vaccenate biosynthesis	0.76
8	PWY-7663	gondoate biosynthesis (anaerobic)	0.74
9	FASYN-ELONG-PWY	fatty acid elongation – saturated	0.74
10	TCA	TCA cycle I (prokaryotic)	0.73
<i>Carbon-fixation pathways</i>			
44	CALVIN-PWY	Calvin-Benson-Bassham cycle	0.59
122	P23-PWY	reductive TCA cycle I	0.41
284	CODH-PWY	reductive acetyl coenzyme A pathway	0.04
304	PWY-5392	reductive TCA cycle II	0.02
360	PWY-5743	3-hydroxypropanoate cycle	0.00
362	PWY-5744	glyoxylate assimilation	0.00
460	PWY-5789	3-hydroxypropanoate/4-hydroxybutanate cycle	0.00

S5.7 Landform, route gradients and core occupancy

Each core site carries one field landform label (Table S14): 44 sand dunes, 9 saline pans, 3 oases and one gravel plain, dune slack, aeolian lake and oil-spill site. Saline pans lie at both ends of the route (three western, one central, five eastern sites) and the oases in the east. Table S15 gives the dune versus saline-pan composition contrast in the site-level CLR coordinates of the route model, with and without adjustment for route position, the route model and route genus correlations refitted on dune sites, and the nine climate-diversity correlations on dune sites. Unadjusted, 16 genera differed between dunes and saline pans (for example *Herpetosiphon*, *Oxalophagus*, *Opitutus* and *Symbiobacterium* higher in dunes, *Thermobaculum*, *Truepera* and *Enterococcus* higher in pans); after route adjustment none passed $q < 0.05$. On dune sites the six temperature and humidity correlations remained supported and the three rainfall correlations lost support.

Tables S17 and S18 list the element loadings of the laboratory-XRF axis and the genera whose site-level CLR values track the axis and site-mean archived-soil pH. Both gradients follow the route (axis $\rho = 0.73$, pH $\rho = 0.82$ with route position), and 104 of the 117 pH-tracking genera are also route-supported, so the three descriptions name one gradient. Table S19 gives genus occupancy per compartment after pooling campaigns within site and subsampling to 12,865 reads. Root-adjacent soil had the largest core (85 genera at $\geq 90\%$ of sites) and the most genera per site; 59 core genera were shared by all three compartments. Genera in the root-adjacent core only included *Brevundimonas*, *Novosphingobium*, *Mesobacillus* and *Terribacillus*; in the surface core only *Kineococcus*, *Conexibacter*, *Cystobacter* and *Nitrospira*; and in the shallow-subsurface core only *Marmoricola*, *Anaeromyxobacter* and *Sporocytophaga*. These occupancy counts describe the pools behind the distance-decay slopes.

Table S14: Field landform of the 60 core sites and site-mean diversity by landform (site means of Shannon diversity, expected richness at 25,000 reads and normalized evenness). Thirds are the western, central and eastern 20 sites by route position. Mann-Whitney tests of dune against saline-pan sites gave $p = 0.35, 0.28$ and 0.93 for the three measures.

Landform	Sites	Thirds	Shannon	Richness	Evenness
sand dune	44	central:18,east:11,west:15	5.77	1635	0.806
saline pan	9	central:1,east:5,west:3	5.15	1259	0.788
desert oasis	3	east:3	4.25	578	0.737
aeolian lake	1	east:1	4.44	759	0.783
dune slack	1	west:1	6.08	1825	0.815
gravel	1	west:1	5.45	1496	0.799
oilspill	1	central:1	6.21	2291	0.811

Table S15: Landform sensitivity of the geographic and climate results. The composition contrast between the 44 dune and 9 saline-pan sites uses the site-level CLR means of the route model, without and with adjustment for route position (linear plus quadratic), 9,999 label permutations. The route model and the genus route correlations were refitted on dune sites only; the nine climate-diversity Spearman correlations are given for all 60 sites and for the 44 dune sites (Benjamini–Hochberg q within each set of nine).

Quantity	All sites / unadjusted	Dune sites / adjusted
Dune vs saline pan, none	pseudo- $F = 4.56$	$p = 0.0013$
Dune vs saline pan, route	pseudo- $F = 1.74$	$p = 0.0362$
linear quadratic		
Route model, all 60 sites	$R^2 = 0.402$	
Route model, 44 dune sites	$R^2 = 0.342$	
Route genera at $q < 0.05$	124 (all) / 71 (dune)	70 in both, same sign
air temperature c vs shannon	$\rho = -0.71$ ($q = 6.02e - 10$)	$\rho = -0.57$ ($q = 0.000139$)
air temperature c vs expected	$\rho = -0.79$	$\rho = -0.71$
richness 25k	($q = 3.12e - 13$)	($q = 5.5e - 07$)
air temperature c vs	$\rho = -0.58$	$\rho = -0.44$ ($q = 0.00448$)
normalized evenness	($q = 1.75e - 06$)	
monthly rain mm vs shannon	$\rho = -0.47$ ($q = 0.000167$)	$\rho = -0.23$ ($q = 0.141$)
monthly rain mm vs expected	$\rho = -0.49$	$\rho = -0.32$ ($q = 0.0439$)
richness 25k	($q = 7.51e - 05$)	
monthly rain mm vs	$\rho = -0.30$ ($q = 0.0206$)	$\rho = -0.11$ ($q = 0.459$)
normalized evenness		
relative humidity pct vs	$\rho = -0.70$	$\rho = -0.57$ ($q = 0.000139$)
shannon	($q = 1.04e - 09$)	
relative humidity pct vs	$\rho = -0.77$	$\rho = -0.69$
expected richness 25k	($q = 2.87e - 12$)	($q = 7.56e - 07$)
relative humidity pct vs	$\rho = -0.52$	$\rho = -0.38$ ($q = 0.0171$)
normalized evenness	($q = 2.89e - 05$)	

Table S16: Genera carrying the compartment differences: the eight largest supported paired mean CLR differences per contrast and direction (295 of 600 tests at Benjamini–Hochberg $q < 0.05$; site bootstrap 95 % interval; sign-flip p over sites). The paired means reproduce the committed displacement loadings of Figure 2d.

Genus	Phylum	Δ CLR	95 % CI	q
<i>shallow-subsurface vs surface: higher in shallow-subsurface</i>				
<i>Polycyclovorans</i>	Pseudomonadota	1.79	[1.24, 2.31]	4.3e-04
<i>Opitutus</i>	Verrucomicrobiota	1.16	[0.83, 1.49]	4.3e-04
<i>Thermosipho</i>	Thermotogota	1.02	[0.58, 1.46]	7.9e-04
<i>Nordella</i>	Pseudomonadota	0.92	[0.50, 1.35]	1.4e-03
<i>Acidiferrimicrobium</i>	Actinomycetota	0.89	[0.51, 1.29]	4.3e-04
<i>MND1</i>	Pseudomonadota	0.89	[0.49, 1.29]	4.3e-04
<i>I-8</i>	Planctomycetota	0.85	[0.38, 1.28]	2.8e-03
<i>Aureibacillus</i>	Bacillota	0.84	[0.42, 1.26]	1.7e-03
<i>shallow-subsurface vs surface: higher in surface</i>				
<i>Planococcus</i>	Bacillota	-1.75	[-2.23, -1.28]	4.3e-04
<i>Nibribacter</i>	Bacteroidota	-1.48	[-1.88, -1.11]	4.3e-04
<i>Kineococcus</i>	Actinomycetota	-1.36	[-1.75, -0.99]	4.3e-04
<i>Deinococcus</i>	Deinococcota	-1.30	[-1.65, -0.97]	4.3e-04
<i>Saccharibacillus</i>	Bacillota	-1.27	[-1.75, -0.81]	4.3e-04
<i>Kineosporia</i>	Actinomycetota	-1.20	[-1.62, -0.80]	4.3e-04
<i>Rufibacter</i>	Bacteroidota	-1.20	[-1.60, -0.78]	4.3e-04
<i>Kocuria</i>	Actinomycetota	-1.13	[-1.50, -0.72]	4.3e-04
<i>root-adjacent vs surface: higher in root-adjacent</i>				
<i>TM7a</i>	Patescibacteria	2.07	[1.50, 2.67]	4.3e-04
<i>Pelagibacterium</i>	Pseudomonadota	1.86	[1.31, 2.44]	4.3e-04
<i>Arsenicitalea</i>	Pseudomonadota	1.67	[1.16, 2.15]	4.3e-04
<i>Ohtaekwangia</i>	Bacteroidota	1.60	[1.06, 2.09]	4.3e-04
<i>Neorhizobium</i>	Pseudomonadota	1.51	[0.98, 2.04]	4.3e-04
<i>DSSF69</i>	Pseudomonadota	1.47	[0.98, 1.93]	4.3e-04
<i>Litchfieldia</i>	Bacillota	1.43	[0.92, 1.99]	4.3e-04
<i>Metabacillus</i>	Bacillota	1.41	[1.10, 1.72]	4.3e-04
<i>root-adjacent vs surface: higher in surface</i>				
<i>Bradyrhizobium</i>	Pseudomonadota	-2.27	[-3.04, -1.52]	4.3e-04
<i>Deinococcus</i>	Deinococcota	-2.01	[-2.48, -1.53]	4.3e-04
<i>Oxalophagus</i>	Bacillota	-1.75	[-2.33, -1.20]	4.3e-04
<i>Blastococcus</i>	Actinomycetota	-1.73	[-2.18, -1.29]	4.3e-04
<i>Valliococcus</i>	Actinomycetota	-1.64	[-2.14, -1.17]	4.3e-04
<i>Rubrobacter</i>	Actinomycetota	-1.58	[-2.01, -1.16]	4.3e-04
<i>Limnobacter</i>	Pseudomonadota	-1.55	[-1.93, -1.15]	4.3e-04
<i>Cystobacter</i>	Myxococcota	-1.54	[-2.01, -1.06]	4.3e-04
<i>root-adjacent vs shallow-subsurface: higher in root-adjacent</i>				
<i>Pelagibacterium</i>	Pseudomonadota	2.37	[1.78, 2.96]	4.3e-04
<i>TM7a</i>	Patescibacteria	2.01	[1.46, 2.57]	4.3e-04
<i>Planococcus</i>	Bacillota	1.99	[1.42, 2.58]	4.3e-04
<i>Brevundimonas</i>	Pseudomonadota	1.95	[1.52, 2.36]	4.3e-04
<i>Devosia</i>	Pseudomonadota	1.95	[1.44, 2.50]	4.3e-04
<i>Pseudomonas</i>	Pseudomonadota	1.87	[1.38, 2.38]	4.3e-04
<i>Neorhizobium</i>	Pseudomonadota	1.76	[1.21, 2.33]	4.3e-04
<i>Nibribacter</i>	Bacteroidota	1.75	[1.40, 2.12]	4.3e-04
<i>root-adjacent vs shallow-subsurface: higher in shallow-subsurface</i>				
<i>Bradyrhizobium</i>	Pseudomonadota	-2.18	[-2.80, -1.61]	4.3e-04
<i>MND1</i>	Pseudomonadota	-2.03	[-2.50, -1.57]	4.3e-04
<i>Nitrospira</i>	Nitrospirota	-2.01	[-2.47, -1.54]	4.3e-04
<i>Thermosipho</i>	Thermotogota	-1.99	[-2.50, -1.45]	4.3e-04
<i>Gaiella</i>	Actinomycetota	-1.94	[-2.42, -1.43]	4.3e-04
<i>Oxalophagus</i>	Bacillota	-1.79	[-2.39, -1.21]	4.3e-04
<i>Symbiobacterium</i>	Bacillota	-1.76	[-2.30, -1.23]	4.3e-04
<i>Rubrobacter</i>	Actinomycetota	-1.63	[-2.09, -1.16]	4.3e-04

Table S17: The laboratory-XRF elemental axis: element loadings of the first principal component after within-campaign standardization, and the genera whose site-level CLR values track the site-mean axis (Spearman over 60 sites; 118 of 200 genera at Benjamini–Hochberg $q < 0.05$). The axis is an elemental axis and does not measure soluble ions.

Element or genus	Phylum	Loading or ρ	q
<i>Element loadings of the laboratory-XRF axis</i>			
Ca		0.413	
Mg		0.406	
Na		0.370	
S		0.354	
Cl		0.338	
Fe		0.284	
Ti		0.281	
K		0.170	
Al		0.158	
P		-0.013	
Si		-0.273	
<i>Genera increasing with the axis (Ca, Mg, Na, S, Cl, Fe, Ti side)</i>			
<i>Polygonibacillus</i>	Bacillota	0.81	1.8e-12
<i>Escherichia-Shigella</i>	Pseudomonadota	0.74	6.2e-10
<i>Sediminibacillus</i>	Bacillota	0.74	6.2e-10
<i>Aquibacillus</i>	Bacillota	0.74	6.2e-10
<i>Halalkalibacter</i>	Bacillota	0.73	1.2e-09
<i>Gracilibacillus</i>	Bacillota	0.72	3.9e-09
<i>Enterococcus</i>	Bacillota	0.71	5.4e-09
<i>Aeromonas</i>	Pseudomonadota	0.69	1.7e-08
<i>Pseudalkalibacillus</i>	Bacillota	0.69	2.5e-08
<i>Crossiella</i>	Actinomycetota	0.65	2.3e-07
<i>Genera decreasing with the axis (Si side)</i>			
<i>Steroidobacter</i>	Pseudomonadota	-0.71	5.4e-09
<i>Caldilinea</i>	Chloroflexota	-0.67	9.3e-08
<i>Phenylobacterium</i>	Pseudomonadota	-0.65	2.7e-07
<i>Paraonexibacter</i>	Actinomycetota	-0.65	3.0e-07
<i>Streptomyces</i>	Actinomycetota	-0.65	3.1e-07
<i>Roseisolibacter</i>	Gemmatimonadota	-0.63	7.1e-07
<i>Gemmatimonas</i>	Gemmatimonadota	-0.63	7.2e-07
<i>Pirellula</i>	Planctomycetota	-0.58	8.9e-06
<i>Blastocatella</i>	Acidobacteriota	-0.58	8.9e-06
<i>Dongia</i>	Pseudomonadota	-0.58	8.9e-06

Table S18: Genera whose site-level CLR values track site-mean archived-soil pH (admitted measurements, 60 sites; 117 of 200 genera at Benjamini–Hochberg $q < 0.05$, 104 of them also supported in the route family).

Genus	Phylum	ρ	q
<i>Increasing with site-mean pH</i>			
<i>Polygonibacillus</i>	Bacillota	0.81	1.1e-12
<i>Sediminibacillus</i>	Bacillota	0.80	3.0e-12
<i>Halalkalibacter</i>	Bacillota	0.78	1.2e-11
<i>Gracilibacillus</i>	Bacillota	0.75	1.6e-10
<i>Halomonas</i>	Pseudomonadota	0.75	2.6e-10
<i>Aquibacillus</i>	Bacillota	0.71	5.2e-09
<i>Aeromonas</i>	Pseudomonadota	0.71	7.4e-09
<i>Ammoniphilus</i>	Bacillota	0.69	3.1e-08
<i>Escherichia-Shigella</i>	Pseudomonadota	0.69	3.1e-08
<i>Truepera</i>	Deinococcota	0.68	4.1e-08
<i>Decreasing with site-mean pH</i>			
<i>Pirellula</i>	Planctomycetota	-0.68	4.4e-08
<i>Caenimonas</i>	Pseudomonadota	-0.68	4.4e-08
<i>Roseisolibacter</i>	Gemmatimonadota	-0.65	2.7e-07
<i>Gemmatimonas</i>	Gemmatimonadota	-0.65	2.7e-07
<i>Steroidobacter</i>	Pseudomonadota	-0.64	3.7e-07
<i>Gaiella</i>	Actinomycetota	-0.64	4.6e-07
<i>Dongia</i>	Pseudomonadota	-0.62	1.2e-06
<i>Adhaeribacter</i>	Bacteroidota	-0.61	1.9e-06
<i>Sphingomonas</i>	Pseudomonadota	-0.60	2.6e-06
<i>Streptomyces</i>	Actinomycetota	-0.60	3.4e-06

Table S19: Genus occupancy per compartment after pooling campaigns within site and subsampling each site to 12,865 reads (20 draws; detected when present in at least half of the draws). Core genera occur at $\geq 90\%$ of sites.

Compartment	Sites	Genera detected	Core genera	Median genera per site	Mean occupancy
surface	60	912	82	274	0.281
shallow-subsurface	60	937	69	284	0.282
root-adjacent	60	1026	85	299	0.283

S6 Phylogenetic diagnostic

We tested whether phylogenetic distance was positively associated with niche difference, a prerequisite for interpreting β NTI/RC-Bray categories, using the same top-800 abundant-ASV set. Each ASV’s niche summary was its abundance-weighted mean on the canonical laboratory-XRF elemental axis. Among the closest 30 % of ASV pairs the association between phylogenetic distance and niche difference was negligible ($r = 0.0262$), and over all pairs it ran in the opposite direction to the requirement (Mantel $r = -0.0837$, $p = 0.002$): the closest distance class had a larger mean niche difference than the farthest (0.808 versus 0.515). The prerequisite failed, so β NTI/RC-Bray assembly-process categories are withheld.

S7 Climate, campaign, rainfall and PMA evidence

A site-clustered Gaussian GEE gave a campaign-by-position interaction for Shannon diversity with Wald $p = 0.00831$ before sequencing-depth adjustment and $p = 0.17476$ after adjustment for log read depth. Campaign-specific shallow-subsurface-surface paired means were +0.434 in Trip 1 (bootstrap 95 % CI 0.241–0.633), +0.216 in Trip 3 ([−0.003,0.433]), −0.212 in Trip 4 ([−0.516,0.066]) and +0.127 in Trip 5 ([−0.286,0.526]); Trip 2 lacked enough complete pairs. The depth-adjusted root-adjacent–shallow contrast remained supported at −0.273 (95 % CI [−0.424,−0.122]), while shallow–surface was direction-only at +0.103 ([−0.012,0.217], $p = 0.0791$). Across the six prespecified extraction models, the root-adjacent–shallow contrast excluded zero in every fit. The root-adjacent–surface direction was stable, but its interval excluded zero in only three fits (additive unadjusted, additive depth-adjusted and protocol-matched pairs) and included zero in the recorded-kit, complete-case and kit-varying-campaign fits.

Extraction kit was recorded for 858 of 1,237 profiles and was partly confounded with campaign. QIACube occurred only in campaigns 4 and 5, which contained no other recorded kit, so its indicator was rank-deficient given campaign among complete cases. Campaigns 1 – −3 contained more than one recorded kit. The log-depth and recorded-kit fits are therefore sensitivities to the recorded batch variables. The campaign-interaction dependence is shown for Shannon diversity. Season, campaign, date and processing context remain inseparable.

S7.1 Long-term climate and collection-weather associations

The field-weather ledger contained 247 quality-controlled primary-site air temperatures (20.0 – −49.4°C), 177 atmospheric-pressure values (894 – −998 mbar) and 239 relative-humidity values (9.5 – −62.5 %). We first averaged the ecological endpoints within site and campaign. Raw pressure was correlated with Shannon diversity, expected richness and normalized evenness ($\rho = -0.570$, -0.591 and -0.349). These associations lost support under the nine-test correction after campaign and a quadratic transect coordinate were included with site-clustered standard errors (all adjusted $q \geq 0.527$). Field temperature and field humidity had no corrected adjusted association. These records are air conditions observed during a site visit.

The long-term analysis used 49 monthly Open-Meteo observations at every core site, from January 2022 through January 2026 [25]. The mean ranges were 26.38 – −30.30°C for temperature, 1.31 – −7.47 mm per month for rain and 24.27 – −35.40 % for relative humidity. Table S20 reports the nine site-level diversity associations. All passed one global Benjamini–Hochberg correction. Expected richness used the available standardized values within each site; all 60 sites remained represented.

Table S20: Site-level Spearman associations between 49-month climate summaries and bacterial diversity. Brackets give 95 % paired-site percentile bootstrap intervals from 9,999 samples. The correction covers all nine tests.

Climate summary	Diversity measure	ρ [95 % interval]	q
Temperature	Shannon	-0.711 [-0.833, -0.520]	6.02×10^{-10}
	Expected richness	-0.795 [-0.887, -0.639]	3.12×10^{-13}
	Evenness	-0.584 [-0.753, -0.359]	1.75×10^{-6}
Rain	Shannon	-0.470 [-0.671, -0.211]	1.67×10^{-4}
	Expected richness	-0.495 [-0.702, -0.226]	7.51×10^{-5}
	Evenness	-0.298 [-0.532, -0.023]	0.0206
Relative humidity	Shannon	-0.701 [-0.834, -0.490]	1.04×10^{-9}
	Expected richness	-0.770 [-0.877, -0.593]	2.87×10^{-12}
	Evenness	-0.522 [-0.713, -0.266]	2.89×10^{-5}

For genus abundance, the analysis selected the 200 genera with the largest mean relative abundance among those present in at least 20 % of profiles. It applied a 0.5 pseudocount, calculated profile-level CLR values and averaged them within site. Across one global family of 600 tests, 112 genera were associated with temperature, 112 with rain and 111 with humidity. The strongest positive temperature associations included *Halalkalibacter* ($\rho = 0.845$) and *Bacillus* ($\rho = 0.833$); the strongest negative associations included *Cellulomonas* and *Marmoricola* (both $\rho = -0.690$). *Truepera* increased with the rain gradient ($\rho = 0.700$), whereas *Massilia* decreased ($\rho = -0.635$).

The climate summaries covaried with geography. Their correlations with the transect coordinate were $\rho = 0.982$ for temperature, 0.551 for rain and 0.919 for humidity. Temperature and humidity were also correlated with one another ($\rho = 0.894$), as were rain and humidity ($\rho = 0.630$). The site-level results therefore describe environmental gradients that coincide with the bacterial geography. Exact site summaries, correlations, genus rankings, field weather models, software versions and checksums are archived under `analysis/v3/environment_associations/`.

S7.2 Fitted short-term rainfall pulse

The primary pulse model used the unfiltered community table and technical means for every available site, expedition and soil position. This gave 617 analysis units: 196 surface, 203 shallow-subsurface and 218 root-adjacent groups. Daily precipitation came from the versioned NASA POWER PRECTOTCORR series [17]; the 60 sites resolved to 20 distinct daily rainfall series. Open-Meteo, which resolved 58 series, provided an independent product sensitivity. We excluded rain on the sampling date because collection times were unavailable.

For each candidate peak from 0.5 to 30 days, the model weighted the preceding 60 complete days with the rise-and-decay curve defined in the main Methods. Expected richness at 25,000 reads, Shannon diversity and normalized evenness formed one search family. The nuisance model included site-by-position baselines, expedition effects, and expedition-specific linear longitude and collection-day trends. The search statistic was the largest positive classical t value across all 60 peaks and three endpoints.

We calibrated this complete search with 19,999 conditional rotations (seed 20260804). Each rotation cyclically moved the 60-day lag axis by one independently sampled amount within each expedition. This preserves the rain amount, rarity, storm clustering and spatial field observed in every expedition, while breaking its timing relative to collection. The calibration therefore holds when storms occur in only some expeditions. A plus-one directional p value tested the positive

pulse hypothesis. A second maximum used absolute t values for a two-sided familywise sensitivity. Site-clustered intervals described the selected effect, and an exact 120-member cyclic and reflected route orbit tested sensitivity to the spatial rain field. We also drew 9,999 multinomial whole-site bootstrap samples (seed 20260805), keeping every observation from one site together. Fixed-curve draws quantified effect and partial- R^2 variation after the fitted nuisance adjustment. Separate draws reselected the peak for the already selected richness endpoint, giving a peak-selection interval. A full-search diagnostic also reselected the endpoint. These intervals are conditional on the fitted nuisance model and one rainfall product; product and route sensitivity remain separate questions.

With NASA POWER, expected richness gave the largest statistic. The curve selected a two-day peak and an estimate of 179.6 expected taxa per millimetre at the kernel peak (site-clustered 95 % CI 92.1 – 267.1). Rain accounted for 3.64 % of the variation remaining after the nuisance model. The complete directional search gave $p = 0.0560$, the two-sided sensitivity gave $p = 0.0787$, and the spatial route orbit gave $p = 0.0833$ in the positive direction. The fitted curve fell to half its peak by 5.36 complete days and to 10 % by 9.78 days. The bootstrap peak-selection interval was 1.5 – 4.5 days; the half-peak and 10 % decay-time intervals were 4.02 – 12.05 and 7.33 – 22.00 days. Open-Meteo selected a one-day peak (directional familywise $p = 0.0432$; two-sided $p = 0.0487$), with a 0.5 – 6.5-day peak-selection interval. Its corresponding decay times were 2.68 and 4.89 days, with much wider intervals of 1.34 – 17.41 and 2.44 – 31.78 days. The products therefore agree on an early association, with delay and decay times spanning these intervals.

When the endpoint was also reselected in each site bootstrap, richness was chosen in 73.7 % of NASA POWER samples and 79.7 % of Open-Meteo samples. Shannon was chosen in 24.7 % and 6.2 %, and evenness in 1.6 % and 14.1 %, respectively. This is selection uncertainty.

Table S21 separates effect, explained-variation and timing uncertainty for the two rainfall products.

Table S21: Effect-size and timing uncertainty for the two rainfall products. Effect intervals use site-clustered sandwich covariance. Partial- R^2 intervals use the fixed-curve site bootstrap. Peak intervals reselect the richness peak in every site bootstrap. The final value gives the frequency with which richness was selected when both endpoint and peak were reselected.

Product	Effect per mm [95 % CI]	Partial R^2 [95 % interval]	Peak, d [95 % interval]	Richness selected
NASA POWER	179.6 [92.1,267.1]	0.0364 [0.0060,0.0787]	2.0 [1.5,4.5]	73.7 %
Open-Meteo	327.5 [141.1,514.0]	0.0384 [0.0057,0.0899]	1.0 [0.5,6.5]	79.7 %

A flexible shape sensitivity fitted 7 disjoint lag intervals together in each rainfall product. For the NASA POWER series, the largest positive estimate occurred at 3–4 days (134.8 expected taxa per millimetre, site-clustered SE 32.4). The Open-Meteo series also placed its largest positive estimate at 3–4 days (77.8, SE 31.7). These jointly fitted coefficients are descriptive shape checks. They support the first several days as the interpretable window. Future rainfall produced no comparable richness pattern in the primary product: its largest richness statistic was 0.055. The largest future statistic instead occurred for evenness at 19.5 days ($t = 2.13$); this negative control was descriptive and sat outside the primary correction.

The selected richness estimate was positive in surface (176.4), shallow-subsurface (228.0) and root-adjacent soil (136.8 expected taxa per millimetre). Contaminant filtering barely changed the estimate (179.4), even though its corrected directional probability crossed the conventional

threshold ($p = 0.0485$). We therefore report the unfiltered estimate. pH adjustment retained the early positive pattern, while a complete paired bulk-soil cohort selected Shannon diversity at two days with weaker evidence. More flexible route terms retained the positive direction but usually weakened the correction (Table S22).

Table S22: Rainfall pulse-response analyses. p_+ is corrected over all peak times and endpoints for the positive pulse hypothesis; $p_{\pm}$ is the two-sided familywise sensitivity. The route-orbit probability tests the spatial rainfall field. Filtering changes the primary probability and leaves the effect estimate unchanged.

Analysis	n	Selected endpoint	Peak (d)	p_+	$p_{\pm}$	Route p_+
NASA POWER, unfiltered	617	Expected richness	2.0	0.0560	0.0787	0.0833
Open-Meteo, unfiltered	617	Expected richness	1.0	0.0432	0.0487	0.0333
Control-filtered	617	Expected richness	2.0	0.0485	0.0711	0.0833
pH-adjusted cohort	547	Expected richness	2.5	0.0480	0.2608	0.0750
Paired bulk soil	179	Shannon diversity	2.0	0.0804	0.2700	0.2417
Quadratic longitude	617	Expected richness	4.5	0.0904	0.4650	0.2333
Quadratic longitude and day	617	Expected richness	2.0	0.0909	0.6763	0.2250
Cubic longitude	617	Expected richness	2.5	0.0478	0.3894	0.1500

For the four complete days before Trip 1 samples, we also compared five gridded products: CHIRPS v2.0, CMORPH V1.0, GPM IMERG Final V07B, NASA POWER and Open-Meteo [8, 23, 10]. Every pair was positively correlated across the 60 sites (Pearson $r = 0.501$ – 0.917 ; Spearman $\rho = 0.554$ – 0.881). This fixed-event comparison supports the spatial exposure pattern, but only NASA POWER and Open-Meteo cover the complete fitted-window sensitivity. Source filenames, official download addresses, byte sizes and checksums are recorded under `data/processed/climate/rain_event_product_sources.tsv`.

Complete cohorts, exposures, scans, nulls, position effects, negative controls, software versions and checksums are archived under `analysis/v3/rain_pulse_response/` and `analysis/v3/rain_pulse_sensitivities/`. Figure S1 shows the primary fitted curve and its calibrated search.

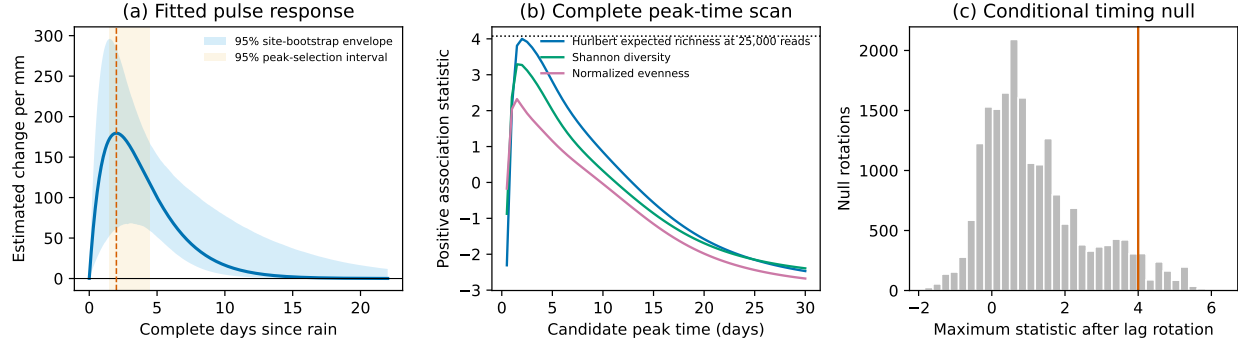


Figure S1: Fitted antecedent-rain pulse. (a) Estimated change in expected richness after one millimetre of rain under the selected NASA POWER pulse kernel. The blue band is the 95 % site-bootstrap curve envelope, the orange band is the 95 % peak-selection interval, and the dashed line marks the two-day fitted peak. (b) Positive-association statistics across all candidate peak times for expected richness, Shannon diversity and normalized evenness; the dotted line is the 95th percentile of the complete-search null. (c) Conditional null distribution of the largest positive statistic across all peak times and endpoints under 19,999 within-expedition lag rotations; the red line marks the observed maximum. The familywise directional probability was 0.056. The curve estimates an association.

S7.3 Paired PMA endpoints

For the PMA pairs, Hurlbert expected richness at the minimum common depth of 123,897 reads was 771.8 untreated versus 608.5 treated (exact paired Wilcoxon $p = 0.00781$; eight of nine pairs lower). The mean treated-minus-untreated difference was -163.3 (95 % paired-aliquot bootstrap interval -274.0 to -69.8). Shannon was 4.19 versus 4.32; its mean difference was 0.130 (95 % interval -0.154 to 0.467 ; $p = 0.652$), so no difference was detected. Both intervals used 99,999 resamples of the nine matched pairs with seed 20260805. The manuscript reports the paired treatment response only. The surviving protocol record omits the PMAxx concentration, dark-incubation time, photoactivation duration and photoactivation device, preventing exact replication of the wet-laboratory treatment. PMA-seq is qualitative in complex communities and is affected by matrix, biomass and taxonomic composition [22]. The staged ASV count table supports re-execution of these endpoints; reproducing it from reads requires separate public raw-read accessions.

S8 Predicted pathway structure and shotgun encoded function

S8.1 PICRUSt2 ecology analysis

PICRUSt2 v2.4.1 predictions matched all 1,227 profiles from the 60 core sites. The unstratified input contained 462 MetaCyc pathways [3]. Sequencing replicates were summed within campaign, site and soil position, yielding 633 grouped profiles. Of the 462 pathways, 416 occurred in at least 20 % of grouped profiles. The primary analysis selected the 200 with the largest mean relative abundance and applied a 0.5 pseudocount before CLR transformation. Analyses of 100 pathways, all 416 prevalent pathways and a 1.0 pseudocount tested the feature-set and zero-treatment choices.

For geography, campaign-by-position means were removed before profiles were averaged within each of the 60 sites. A pre-specified linear and quadratic transect model accounted for 23.33 % of predicted pathway composition (95 % delete-one-site jackknife interval, 15.37–31.28 %; $p = 0.0001$;

9,999 whole-site permutations). Across pathway-set, pseudocount and leave-one-campaign analyses, the explained share ranged from 21.31 to 36.14 %, and every permutation test gave $p = 0.0001$.

For soil position, 172 campaign-by-site blocks contained surface, shallow-subsurface and root-adjacent profiles. Each complete block was centred on its own three-position mean and then averaged within site, making the 60 sites the independent units. The omnibus position statistic was pseudo- $F = 6.37$ ($p = 0.0001$). Standardized paired displacements were 0.453 for root-adjacent minus shallow-subsurface ($q = 0.00015$), 0.429 for root-adjacent minus surface ($q = 0.00015$), and 0.330 for shallow-subsurface minus surface ($q = 0.0194$). The omnibus result and both root-adjacent contrasts passed every pathway-set, pseudocount and leave-one-campaign test. The shallow-subsurface-surface contrast gave $p = 0.136$ when Trip 3 was omitted, so it is the weaker pairwise result.

Pathway-wise sign-flip tests retained the same site-level paired design. One global correction covered 600 pathway-by-position tests: 121 root-adjacent-shallow, 73 root-adjacent-surface and 76 shallow-surface associations passed $q < 0.05$. The supported predictions included amino-acid and cofactor biosynthesis, carbohydrate degradation and cell-envelope pathways. A separate family of 200 pathway-transect correlations retained 92 pathways. These follow-up tests identify predicted pathways for direct measurement.

Several supported predictions motivate more specific follow-up experiments. Along the west-east route, *cis*-vaccenate biosynthesis, gondoate biosynthesis and saturated fatty-acid elongation each had $\rho = 0.610$ – 0.616 ($q \leq 3.24 \times 10^{-6}$), whereas palmitate biosynthesis had $\rho = -0.527$ ($q = 1.18 \times 10^{-4}$). Root-adjacent soil exceeded both bulk-soil positions for predicted glucose and xylose degradation, fructuronate and 4-aminobutanoate degradation, polyamine biosynthesis and thiamin diphosphate biosynthesis (all $q \leq 0.0162$). These are pathway predictions. The selected MetaCyc set contains no dedicated UV-repair pathway, so testing UV protection or DNA repair requires a pre-specified gene-level analysis. The complete 600 position tests and 200 geographic tests are in `analysis/v3/picrust2_ecology/`.

The descriptive taxon-context module grouped the 462 pathways by keywords in their MetaCyc descriptions (Table S12). Biosynthesis pathways held 66.8 % of mean predicted pathway abundance, energy and central metabolism 17.2 %, degradation or utilization 10.7 % and other pathways 5.3 %. Aerobic respiration I (cytochrome *c*) was the most abundant pathway (1.70 %); the Calvin-Benson-Bassham cycle ranked 44th (0.59 %) and the reductive TCA cycle I 122nd (0.41 %), and the PICRUSt2 MetaCyc set contains no light-reaction pathway (Table S13). Of the 92 supported route correlations, 63 were biosynthesis pathways (54 increasing eastward); of the 270 supported compartment contrasts, 208 were biosynthesis pathways. The classes are keyword groupings. Table S24 lists the five largest supported pathway contrasts per compartment pair and direction.

Weighted NSTI had median 0.1648 across the 1,227 profiles (range 0.0053 – -0.7819). The median was 0.2011 in shallow-subsurface, 0.1681 in surface and 0.1340 in root-adjacent profiles. Lower NSTI means that the contributing ASVs are closer to the reference genomes used for prediction. Exact grouped cohorts, primary and sensitivity tests, pathway-level results, NSTI summaries, seeds and checksums are archived under `analysis/v3/picrust2_ecology/`.

S8.2 Resolution-matched shotgun encoded-function test

Table S23 reports the resolution-matched functional turnover test and its intact-genome-label null.

Table S23: Resolution-matched encoded-function null. “Encoded” is the median Bray–Curtis dissimilarity of abundance-weighted KO profiles; the null reassigns intact genome annotation profiles among fixed genome-abundance vectors (999 permutations). p_L and p_U are the plus-one-corrected lower- and upper-tail values.

Annotation basis	Group	n	Taxonomic	Encoded	Encoded/taxonomic	Null median [95 % interval]	p_L	p_U
KO copy count	All	150	0.82272	0.13386	0.16271	0.08733 [0.08459, 0.09067]	1.000	0.001
	shallow-subsurface	44	0.80929	0.13270	0.16397	0.08400 [0.08029, 0.08838]	1.000	0.001
	Root-adjacent	79	0.81810	0.13348	0.16316	0.08889 [0.08567, 0.09269]	1.000	0.001
	Surface	27	0.75209	0.11254	0.14964	0.08138 [0.07595, 0.08765]	1.000	0.001
Within-genome relative	All	150	0.82272	0.13659	0.16602	0.08882 [0.08592, 0.09315]	1.000	0.001
	shallow-subsurface	44	0.80929	0.13615	0.16824	0.08544 [0.08165, 0.09040]	1.000	0.001
	Root-adjacent	79	0.81810	0.13418	0.16402	0.09041 [0.08702, 0.09523]	1.000	0.001
	Surface	27	0.75209	0.11573	0.15388	0.08284 [0.07758, 0.09023]	1.000	0.001

The staged, checksum-tracked CoverM and eggNOG-derived inputs support re-execution of the following downstream test; regeneration from raw reads additionally requires the separate MAG catalogue and upstream metadata. The observed encoded-functional dissimilarity exceeded the complete 95 % null interval under both annotation bases and in every compartment. Accordingly, encoded-functional turnover was lower than taxonomic turnover in absolute terms, but higher than the resolution-matched genome-label null. Observed turnover exceeded every null replicate ($p_U = 0.001$). The null preserves the genome-abundance matrix, every intact annotation profile and KO prevalence. On average 0.28465 of the CoverM total was assigned to the annotated 990-genome set, so the result concerns genome-resolved encoded potential in that fraction.

A top- K sensitivity retained 1,000, 2,000 or 5,000 KOs. Observed median KO-copy Bray–Curtis dissimilarity was 0.1005, 0.1339 and 0.1721, compared with label-null medians of 0.0670, 0.0873 and 0.1174, respectively. Observed turnover exceeded the null upper interval in all three configurations ($p_U = 0.001$ throughout), so the conclusion holds across the K choices.

Across 125 samples and 3,471 shared KOs, the PICRUSt2-shotgun whole-profile Spearman correlation had median 0.743 (95 % whole-site bootstrap interval 0.731 – –0.751) and community-mean 0.806 (0.793 – –0.812). The 9,999 bootstrap samples kept all matched profiles from the same field site together and used seed 20260805. Rare-marker agreement is much weaker (RuBisCO $r = 0.16$, two-sided $p = 0.067$), so the agreement is a whole-profile result. These predictions used separately executed Trips 1 – –4 and Trip 5 input sets containing 330,830 ASVs; the primary ecological analyses used the 351,472-ASV combined canonical table.

The staged positive-only metabolic-marker table contains 132 genome records annotated with both RuBisCO and phosphoribulokinase. The complete upstream screened-genome manifest was not preserved in the local analysis bundle, so the table reports counts only. This supports candidate Calvin-cycle genomic potential in the recovered output. The supplied pathway screen contains no positive calls for rTCA, Wood–Ljungdahl or the 3-HP cycle. It contains 33 single-marker hits for the hydroxybutyrate cycle, a marker shared with other pathways. Hydrogenase and CO-dehydrogenase product labels identify candidate genes for high-affinity atmospheric oxidation; *mcrA* and *nifH* non-detections are bounded by recovery and annotation sensitivity. No profile-HMM, catalytic-motif, phylogenetic, synteny or direct-read confirmation was included, so no trace-gas pathway or activity claim is made.

S8.3 Trait-gene screen

For a fixed list of 20 trait genes (trace-gas oxidation, carbon fixation, nitrogen cycling, osmolytes, pigments, sporulation, DNA repair and oxidative-stress defence; KO membership in `analysis/v3/trait_genes/trait_definitions.tsv`) we report two gene sources separately. Table S25 gives, for

the 975 annotated genomes of the companion shotgun catalogue, the fraction of genomes carrying each trait, the abundance-weighted share of the recruited community carrying it in the 150 shotgun libraries by compartment, and the leading carrier phylum. Table S26 gives the PICRUSt2 predicted share of KO abundance by compartment, the paired site-level fold changes between compartments, the route correlation, and the agreement between the two sources in the 119 matched libraries. The two sources agree best for the trace-gas ($\rho = 0.71$ and 0.74), Dps (0.76), superoxide dismutase (0.66), ectoine (0.65), catalase (0.57), trehalose *otsA/otsB* (0.53) and sporulation (0.51) traits and poorly for genes that are near universal (*uvrA*, *recA*) or rare (*nifH*, *psbA*, *nosZ*). The catalogue holds 59 Bacillota genomes against 345 Actinomycetota, so it under-represents the endospore formers that supply 22.5% of amplicon reads; its compartment shares are descriptive and most libraries are root-adjacent. The hydrogenase trait pools group 1 KOs (*hyaB*, *hhyL*, *hoxH*). Gene presence and predicted share describe potential.

Table S24: Leading supported predicted-pathway contrasts per compartment pair and direction (five largest paired mean CLR differences among the 270 supported tests of the 600-test family).

Pathway	Description	Δ CLR
<i>shallow-subsurface-surface: higher in shallow-subsurface</i>		
PWY-5104	L-isoleucine biosynthesis IV	0.22
P42-PWY	incomplete reductive TCA cycle	0.15
PWY-6609	adenine and adenosine salvage III	0.14
PWY-5121	superpathway of geranylgeranyl diphosphate biosynthesis II (via MEP)	0.12
PWY-5659	GDP-mannose biosynthesis	0.09
<i>shallow-subsurface-surface: higher in surface</i>		
PWY0-1061	superpathway of L-alanine biosynthesis	-0.36
PWY0-1479	tRNA processing	-0.35
PWY-6519	8-amino-7-oxononanoate biosynthesis I	-0.33
BIOTIN-BIOSYNTHESIS-PWY	biotin biosynthesis I	-0.31
PWY-7254	TCA cycle VII (acetate-producers)	-0.24
<i>root-adjacent-shallow-subsurface: higher in shallow-subsurface</i>		
PWY-5104	L-isoleucine biosynthesis IV	-0.38
PWY-5659	GDP-mannose biosynthesis	-0.19
PWY-6467	Kdo transfer to lipid IVA III (Chlamydia)	-0.19
GLYCOCAT-PWY	glycogen degradation I (bacterial)	-0.17
DENOVOPURINE2-PWY	superpathway of purine nucleotides de novo biosynthesis II	-0.17
<i>root-adjacent-shallow-subsurface: higher in root-adjacent</i>		
PWY-1861	formaldehyde assimilation II (RuMP Cycle)	0.53
PWY-5971	palmitate biosynthesis II (bacteria and plants)	0.49
PWY-2941	L-lysine biosynthesis II	0.37
PWY-7254	TCA cycle VII (acetate-producers)	0.35
PWY0-1061	superpathway of L-alanine biosynthesis	0.31
<i>root-adjacent-surface: higher in root-adjacent</i>		
PWY-1861	formaldehyde assimilation II (RuMP Cycle)	0.51
PWY-5837	1,4-dihydroxy-2-naphthoate biosynthesis I	0.31
PWY-5863	superpathway of phyloquinol biosynthesis	0.30
PWY-2941	L-lysine biosynthesis II	0.28
PWY-5897	superpathway of menaquinol-11 biosynthesis	0.27
<i>root-adjacent-surface: higher in surface</i>		
PWY-6467	Kdo transfer to lipid IVA III (Chlamydia)	-0.17
PWY-5104	L-isoleucine biosynthesis IV	-0.16
PWY-6737	starch degradation V	-0.16
PWY-1269	CMP-3-deoxy-D-manno-octulosonate biosynthesis I	-0.15
PWY0-1479	tRNA processing	-0.14

Table S25: Trait genes in the 975 annotated genomes of the companion shotgun catalogue (KO membership in `analysis/v3/trait_genes/trait_definitions.tsv`): fraction of genomes carrying at least one member KO, the abundance-weighted share of the recruited community carrying the trait in the 150 shotgun libraries by compartment (surface 27, shallow-subsurface 44, root-adjacent 79 libraries), and the leading carrier phylum with its genome count.

Trait	Category	Genomes (%)	Surface	Shallow	Root-adj.	Leading carriers
aerobic CO dehydrogenase large subunit (coxL)	trace gas	38.5	43.8	45.5	40.4	Actinomycetota 170
[NiFe]-hydrogenase large subunit (group 1/1h KOs hyaB, hhyL, hoxH)	trace gas	22.6	17.2	31.9	15.0	Actinomycetota 126
RuBisCO large subunit (rbcL/cbbL)	carbon fixation	24.6	18.2	24.1	18.1	Actinomycetota 90
photosystem II D1 protein (psbA)	carbon fixation	0.2	0.1	0.2	0.1	Pseudomonadota 1
nitrogenase iron protein (nifH)	nitrogen	1.0	0.5	0.3	0.8	Pseudomonadota 3
ammonia/methane monooxygenase subunit A (amoA/pmoA)	nitrogen	3.3	3.3	4.7	1.9	Actinomycetota 16
dissimilatory nitrite reductase (nirK, nirS)	nitrogen	32.4	30.6	38.8	23.1	Actinomycetota 149
nitrous-oxide reductase (nosZ)	nitrogen	4.1	5.1	4.0	2.2	Pseudomonadota 14
urease alpha subunit (ureC)	nitrogen	31.2	24.0	27.9	35.3	Pseudomonadota 132
trehalose synthesis (otsA, otsB)	osmolyte	72.7	85.6	89.0	83.8	Actinomycetota 336
trehalose from maltooligosaccharides (treY, treZ)	osmolyte	70.1	74.3	79.1	76.5	Actinomycetota 258
ectoine synthesis (ectA, ectB, ectC)	osmolyte	36.4	25.9	28.0	31.3	Actinomycetota 145
phytoene synthase (crtB), carotenoid pigments	pigment	49.3	60.6	52.7	53.4	Pseudomonadota 199
sporulation master regulator (spo0A)	dormancy	7.3	5.1	4.3	7.5	Bacillota 41
deoxyribodipyrimidine photolyase (phrB)	DNA repair	23.6	34.5	17.8	26.0	Pseudomonadota 71
UvrABC system protein A (uvrA)	DNA repair	94.6	93.2	92.0	93.8	Actinomycetota 324
recombinase A (recA)	DNA repair	92.2	88.9	90.9	92.4	Actinomycetota 324
catalase (katE) and catalase-peroxidase (katG)	oxidative stress	77.4	86.7	76.3	84.1	Actinomycetota 266
Fe/Mn superoxide dismutase (sodA)	oxidative stress	82.8	68.4	67.4	78.8	Pseudomonadota 238
DNA protection during starvation protein (dps)	oxidative stress	65.1	58.2	47.3	69.5	Actinomycetota 177

Table S26: PICRUSt2 predicted trait-gene share of KO abundance ($\times 10^4$) by compartment; paired site-level fold changes between compartments (sign-flip tests, Benjamini–Hochberg over 60 tests; * $q < 0.05$); Spearman correlation of site means with route position (* $q < 0.05$ over 20 tests); and the Spearman agreement between the genome-derived and predicted shares in the 119 matched shotgun libraries.

Trait	Surface	Shallow	Root	Sh/Su	Ro/Su	Ro/Sh	Route ρ	Agreement
coxL CO dehydrogenase	3.89	4.26	3.57	1.09*	0.78*	0.74*	-0.11	+0.71
NiFe hydrogenase large	1.68	1.90	1.36	1.16*	0.77*	0.67*	+0.05	+0.74
rbcL RuBisCO	1.23	1.20	1.04	1.01	0.81*	0.82*	+0.26	+0.33
psbA photosystem II	0.03	0.02	0.03	1.25	0.90	0.81	+0.05	+0.18
nifH nitrogenase	0.48	0.53	0.41	1.19	1.19	1.04	+0.08	+0.04
amoA pmoA ammonia monooxygenase	0.05	0.03	0.03	1.33	1.36	1.21	-0.24	+0.41
nirK nirS nitrite reductase	1.32	1.28	1.14	0.96	0.88*	0.93	-0.23	+0.40
nosZ N2O reductase	0.84	0.82	0.80	1.01	0.95	0.95	+0.20	-0.10
ureC urease	2.80	2.60	2.74	0.92*	0.98	1.08*	-0.27	+0.26
otsA otsB trehalose	5.06	5.15	4.51	1.01	0.86*	0.87*	-0.71*	+0.53
treY treZ trehalose	4.38	4.34	3.74	0.95	0.84*	0.91	-0.67*	+0.40
ectABC ectoine	1.75	2.24	2.74	1.21*	1.22*	1.05	+0.36*	+0.65
crtB phytoene synthase	4.00	3.92	3.67	0.97	0.92*	0.94*	-0.59*	+0.13
spo0A sporulation	1.05	1.04	1.32	0.91	1.41*	1.49*	+0.56*	+0.51
phrB photolyase	4.05	4.16	3.88	1.03	0.94*	0.93*	-0.27	-0.14
uvrA excision repair	7.62	7.89	7.51	1.03*	0.99	0.96*	-0.24	-0.21
recA recombination repair	5.59	5.80	5.53	1.04*	1.00	0.95*	-0.57*	+0.06
katE katG catalase	9.37	8.50	9.74	0.91*	1.09*	1.20*	-0.16	+0.57
sodA superoxide dismutase	5.96	5.97	6.34	1.00	1.06*	1.06*	+0.70*	+0.66
dps DNA protection	4.54	4.40	4.75	0.97*	1.07*	1.10*	+0.43*	+0.76

S9 Cross-desert and regional context

S9.1 Quantitative comparison with public Atacama surveys

We compared effects across surveys because they used different 16S regions, reference databases and sampling designs. The Atacama aridity-gradient input is an archived site-level table derived from QIITA study 10360. The upstream V4 OTU profiles were rarefied 100 times to 12,000 reads and technical replicates were averaged within site. The replayed analysis retained 16 sites with complete soil relative humidity and spatial information. Soil relative humidity was positively associated with Shannon diversity ($\rho = 0.8904$, $p = 3.80 \times 10^{-6}$). The association remained after ranked humidity and diversity were adjusted for latitude, longitude and elevation (partial $\rho = 0.6753$, $p = 0.0113$). Rarefied richness gave the same direction (raw $\rho = 0.8712$; partial $\rho = 0.6445$).

For the Atacama pit, all 62 reprocessed intracellular-DNA profiles exceeded 8,000 reads. Replicates represented 23 sampled depths from 2.5 to 420 cm. We combined replicate counts within depth, retained the 200 leading genera and tested three zones in CLR coordinates. Composition differed among zones (pseudo- $F = 1.463$, $p = 0.0014$, 9,999 label permutations). After 100 rarefactions per profile and averaging within depth, richness increased only weakly with depth ($\rho = 0.381$, $p = 0.0725$), and Shannon diversity had no monotonic trend ($\rho = 0.120$, $p = 0.587$). The compositional result supports depth as a recurring source of habitat structure; the diversity direction depends on the depth design.

Soil relative humidity in the Atacama and a 49-month atmospheric humidity mean in the Rub' al-Khali (Empty Quarter) are different exposures. The Atacama pit is one vertical profile sampled to 4.2 m, whereas the Empty Quarter compartments span only 0–10 cm and include root-adjacent soil. Exact inputs, parameters, software versions and checksums are recorded under `analysis/v3/cross_desert_context/`. The archived Atacama-gradient table reproduces the site-

level tests, but regeneration of that table from the original QIITA BIOM file remains an upstream step.

S9.2 Genus-set overlap with the Atacama pit

To place the within-transect replacement against a between-desert difference under one detection rule, every Empty Quarter core profile and every Atacama pit profile (PRJEB39249; Bacteria and Archaea only) was subsampled 20 times to 8,000 reads (seed 20260903), and a genus counted as detected in a set when it occurred in at least 10 % of subsampled profiles. Table S11 gives the overlaps. The western and eastern thirds of the transect shared 267 of 418 detected genera (Jaccard 0.639), the adjacent thirds 0.825 and 0.677, and every one of the 50 leading genera of one end was detected at the other. The Empty Quarter and the pit shared 43 of 397 genera (Jaccard 0.108; 0.091 for Empty Quarter surface soil against the 2.5–10 cm pit profiles); 17 of the 50 leading Empty Quarter genera and 31 of the 50 leading pit genera were detected in the other survey. The pit used V4 primers on intracellular DNA and sampled one vertical profile, so the between-desert difference is inflated by method; the comparison keeps the feature tables separate.

S9.3 Design boundaries in related studies

Table S27 places the survey against regional observational and experimental precedents while retaining their design boundaries.

Table S27: Studies that define the comparison and evidential boundary.

Study	Design	Relevance to this manuscript
McKay et al. 2016, “An Unusual Inverted Saline Microbial Mat Community in an Interdune Sabkha in the Rub’ al Khali” [15]	One groundwater-fed Liwa sabkha, UAE; salt mat microbiology and geochemistry	Direct Empty Quarter precedent, a specialized saline mat
Yasir et al. 2015, “Composition of soil microbiome along elevation gradients in southwestern highlands of Saudi Arabia” [24]	Ten soils across four southwestern highland elevations; 16S V6 pyrosequencing and soil chemistry	Early Saudi spatial soil-microbiome precedent; a small highland gradient rather than a repeated hyperarid transect
Aslam et al. 2016, “Soil compartment is a major determinant of the impact of simulated rainfall on desert microbiota” [1]	In-situ watering of Saudi desert crust, root-attached and broader root-zone soil	Direct experimental compartment/rainfall comparator; a manipulated response trajectory complementary to our five observational dates
Eida et al. 2018, “Desert plant bacteria reveal host influence and beneficial plant growth properties” [5]	Barren soil, rhizosphere and root endosphere of identified hosts in Jizan and Al Wahbah	Defines the sampling and host metadata needed for rhizosphere/endosphere and host-filter claims

Study	Design	Relevance to this manuscript
Khan and Khan 2020, “Microbial communities and their predictive functional profiles in the arid soil of Saudi Arabia” [12]	Abha, Hafar Al Batin and Muzahmiya; 16S and PICRUSt; includes rhizosphere	Direct Saudi arid-soil precedent; prevents a “first Saudi survey” claim
Maurice et al. 2023, “Fertility islands, keys to the establishment of plant and microbial diversity in a highly alkaline hot desert” [14]	AIUla vegetation patches, soils, geochemistry and microbial diversity	Saudi fertility-island and plant-associated geochemistry comparator; unlike operationally defined root-adjacent soil
Maurice et al. 2024, “Networking the desert plant microbiome” [13]	Roots and rhizospheres of six identified plant species in AIUla, Saudi Arabia	Shows the host and root metadata needed for a plant-selective-filter mechanism
Jalal et al. 2022, Saudi coastal halophyte compartments [11]	Southern Corniche halophyte-associated soil, rhizosphere, root and leaf endosphere samples	Saudi saline-soil precedent; a specialized host system
Mousa et al. 2024, “High-throughput sequencing reveals the structure and metabolic resilience of desert microbiome confronting climate change” [16]	Rhizosphere and endosphere sequencing of native Arabian desert plants in Dubai and Al Ain, UAE	UAE/Arabian plant-desert comparator; distinct from the open-soil landscape design used here
Pan et al. 2022, “Aridity Threshold Induces Abrupt Change of Soil Abundant and Rare Bacterial Biogeography” [18]	Chinese dryland gradient spanning aridity threshold; abundant/rare assembly	Hypothesis-level comparison; an aridity response requires a crossed design beyond our one-dimensional transect
Bay et al. 2020, Negev soil biogeography [2]	99 soils across as much as 189 km; fine-resolution community turnover and distance decay	Independent hot-desert support for spatial turnover; primer, distance metric and landscape differ, so the comparison is qualitative
Ronca et al. 2015, Namib dune/interdune transects [20]	Surface soils across repeated dune and interdune habitats	Shows habitat-specific turnover in another sand desert
Pombubpa et al. 2020, Mojave biocrust survey [19]	48 surface and subsurface profiles across four sites and crust types	Independent support for depth-related composition; biocrust and open-sand positions are different habitats
Demergasso et al. 2023, “Hyperarid soil microbial community response to simulated rainfall” [4]	Controlled, replicated 30-day rainfall treatment	Shows the trajectory and controls needed to establish bloom/relaxation

Study	Design	Relevance to this manuscript
Horstmann et al. 2024, “Persistent microbial communities in hyperarid subsurface habitats of the Atacama Desert” [9]	Depth profile to 4.2 m; intracellular DNA	Appropriate subsurface comparison for our 5 – 10 cm samples
El Moatassem et al. 2025, “Soil salinity hazard monitoring with portable X-ray fluorescence spectrometry and electrical conductivity meters” [6]	pXRF models calibrated against EC	Establishes the independent validation required before calling the XRF axis salinity
Tribbia et al. 2026, “Trace gas oxidation supports sub-surface microbial communities across Namib Desert fog and aridity gradients” [21]	Gas uptake, wetting, ATP, respiration, abundance and geochemistry	Defines the direct activity evidence missing from our marker inventory

This study is a large, repeated landscape survey of open sand, shallow subsurface and root-adjacent Saudi Empty Quarter soils, joined to distinct field/laboratory XRF products and an analysis-ready shotgun subset.

S10 Uncertainty estimates for headline results

Table S28 lists the point estimates that carry the main ecological interpretation. Brackets are 95 % sampling intervals unless they are identified as a sensitivity range or interquartile range. These quantities answer different questions. Sampling intervals describe variation among sites or matched pairs. Rain peak-selection intervals include the choice among candidate peak times but remain conditional on one rainfall product and the fitted nuisance adjustment. The pH influence and XRF model ranges describe specific analysis choices. The contamination interquartile range describes variation among linked profiles.

For paired contrasts and correlations, bootstrap resampling kept repeated observations from one site or aliquot pair together. For geographic R^2 , distance slopes and replacement shares, the delete-one-site jackknife removed the complete site; for pairwise distances it also removed every pair involving that site. Functional-profile bootstraps kept all profiles from a field site together. Exact machine-readable values, method labels, independent units and source files are in `analysis/v3/headline_uncertainty/headline_uncertainty.tsv`.

Table S28: Headline estimates and quantified uncertainty. Bootstrap intervals are percentile intervals. CI denotes a 95 % confidence interval.

Estimand	Estimate [interval or range]	Basis and independent unit
Geography		
Quadratic transect partial R^2	0.401 [0.324,0.477]	Site-jackknife CI; 60 sites
Aitchison slope per 100 km, surface	1.349 [0.903,1.795]	Site-jackknife CI; 60 sites
Aitchison slope per 100 km, shallow-subsurface	1.507 [1.089,1.925]	Site-jackknife CI; 60 sites
Aitchison slope per 100 km, root-adjacent	1.175 [0.695,1.656]	Site-jackknife CI; 60 sites
Replacement share, surface	0.715 [0.636,0.793]	Site-jackknife CI; 60 sites
Replacement share, shallow-subsurface	0.692 [0.607,0.776]	Site-jackknife CI; 60 sites
Replacement share, root-adjacent	0.747 [0.676,0.818]	Site-jackknife CI; 60 sites
Soil position		
Standardized composition, shallow minus surface	0.331 [0.306,0.404]	Paired-site bootstrap; 60 sites
Standardized composition, root-adjacent minus surface	0.433 [0.386,0.509]	Paired-site bootstrap; 60 sites
Standardized composition, root-adjacent minus shallow	0.478 [0.423,0.556]	Paired-site bootstrap; 60 sites
Mean Shannon difference, shallow minus surface	0.075 [−0.089,0.235]	Paired-site bootstrap; 60 sites
Mean Shannon difference, root-adjacent minus surface	−0.130 [−0.321,0.079]	Paired-site bootstrap; 60 sites
Mean Shannon difference, root-adjacent minus shallow	−0.206 [−0.398,−0.001]	Paired-site bootstrap; 60 sites
Mean evenness difference, shallow minus surface	0.0125 [0.0008,0.0251]	Paired-site bootstrap; 60 sites
Mean evenness difference, root-adjacent minus surface	−0.0315 [−0.0448,−0.0186]	Paired-site bootstrap; 60 sites
Mean evenness difference, root-adjacent minus shallow	−0.0440 [−0.0572,−0.0313]	Paired-site bootstrap; 60 sites
Long-term climate		
Temperature versus Shannon, Spearman ρ	−0.711 [−0.833,−0.520]	Paired-site bootstrap; 60 sites
Temperature versus expected richness, ρ	−0.795 [−0.887,−0.639]	Paired-site bootstrap; 60 sites
Temperature versus evenness, ρ	−0.584 [−0.753,−0.359]	Paired-site bootstrap; 60 sites
Monthly rain versus Shannon, ρ	−0.470 [−0.671,−0.211]	Paired-site bootstrap; 60 sites
Monthly rain versus expected richness, ρ	−0.495 [−0.702,−0.226]	Paired-site bootstrap; 60 sites
Monthly rain versus evenness, ρ	−0.298 [−0.532,−0.023]	Paired-site bootstrap; 60 sites
Humidity versus Shannon, ρ	−0.701 [−0.834,−0.490]	Paired-site bootstrap; 60 sites
Humidity versus expected richness, ρ	−0.770 [−0.877,−0.593]	Paired-site bootstrap; 60 sites
Humidity versus evenness, ρ	−0.522 [−0.713,−0.266]	Paired-site bootstrap; 60 sites
pH and elemental composition		
Geographic partial R^2 before pH adjustment	0.358 [0.358,0.362]	Prespecified influence range
Geographic partial R^2 after pH adjustment	0.196 [0.187,0.196]	Prespecified influence range
pH transect partial R^2	0.846 [0.826,0.846]	Prespecified influence range
Elemental-axis partial R^2	0.00358 [0.00311,0.00543]	Prespecified model range
Recent rain		
NASA POWER richness change per mm at peak	179.6 [92.1,267.1]	Site-clustered sandwich CI; 60 sites
NASA POWER partial R^2 at peak	0.0364 [0.0060,0.0787]	Fixed-curve site bootstrap; 60 sites
NASA POWER selected peak, complete days	2.0 [1.5,4.5]	Peak-selection site bootstrap; 60 sites
Open-Meteo richness change per mm at peak	327.5 [141.1,514.0]	Site-clustered sandwich CI; 60 sites
Open-Meteo partial R^2 at peak	0.0384 [0.0057,0.0899]	Fixed-curve site bootstrap; 60 sites
Open-Meteo selected peak, complete days	1.0 [0.5,6.5]	Peak-selection site bootstrap; 60 sites
Predicted function		
PICRUSt2 pathway transect R^2	0.234 [0.155,0.314]	Site-jackknife CI; 60 sites
PICRUSt2 position, shallow minus surface	0.330 [0.296,0.501]	Paired-site bootstrap; 60 sites
PICRUSt2 position, root-adjacent minus surface	0.429 [0.350,0.566]	Paired-site bootstrap; 60 sites
PICRUSt2 position, root-adjacent minus shallow	0.453 [0.412,0.549]	Paired-site bootstrap; 60 sites
Median within-sample PICRUSt2-shotgun ρ	0.743 [0.731,0.751]	Whole-site bootstrap; 125 profiles
Community-mean PICRUSt2-shotgun ρ	0.806 [0.793,0.812]	Whole-site bootstrap; 125 profiles
Assay controls		

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Table S28 continued

Estimand	Estimate [interval or range]	Basis and independent unit
PMA richness difference, treated minus untreated	-163.3 [-274.0,-69.8]	Paired-aliquot bootstrap; nine pairs
PMA Shannon difference, treated minus untreated	0.130 [-0.154,0.467]	Paired-aliquot bootstrap; nine pairs
Candidate-read fraction per linked profile	0.00403 [0.00155,0.00987]	Empirical median [IQR]; 217 profiles

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